nature communications

文章

https://doi.org/10.1038/s41467-025-65690-3

用于药物诱导肝损伤预测的大规模人类毒理基因组学资源

收到：2025年1月8日

接受日期：2025年10月21日

Published online: 13 November 2025

检查更新

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药物性肝损伤（DILI）仍然是药物开发中最关键的挑战之一，导致患者安全问题、临床试验失败和药物撤回。我们介绍了ToxPredictor，一种毒理基因组学框架，将来自人原代肝细胞的RNA测序数据与药代动力学数据相结合，以预测剂量分辨的DILI风险和安全边界。其核心是DILImap，这是一个为DILI研究定制的RNA测序文库，包含多种浓度下的300种化合物。在盲法验证中，ToxPredictor在100%特异性下实现了88%的敏感性，优于最先进的方法。它识别了近期的III期临床失败案例，包括Evobrutinib、TAK-875和BMS-986142，这些在动物实验中被忽视。除了预测功能，ToxPredictor还提供了对肝毒性通路的机制性洞察，使早期风险降低和安全决策可执行。与单终点读数——即使是来自3D模型的——不同，转录组学提供了多维系统层面的视角。

药物性肝损伤（DILI）在药物开发中呈现一个尚不完全理解的晚期挑战，每家制药公司每年估计损失3.5亿美元[1]。其在临床人群中的稀有性和不可预测性，通常少于每1万人中1例，阻碍了临床研究中的检测，使其严重性在上市后才得以显现。动物模型未能识别约一半在临床中表现出DILI的药物[2]。这使得DILI成为药物候选失败和市场撤回的主要原因，阻碍了新疗法的发展[3]。

DILI 由复杂的、多因素机制引起，涉及剂量依赖的内在机制，并且当前方法无法预测受遗传影响的特异反应[4,5]易感性、环境和个体健康状况[6]。药物性肝损伤（DILI）涉及多种细胞功能紊乱，包括线粒体功能障碍、氧化应激、胆汁酸失衡、特定酶或转运蛋白的抑制以及反应性代谢物的形成[7,8]。然而，其确切机制和影响因素尚未完全阐明[4]，而且缺乏生物标志物阻碍了早期检测[9]。临床前方法，如定量构效关系（QSAR）模型，具有特异性低和二元预测的特点，缺乏机制性见解[10,11]。体外模型使用多种细胞来源（例如 HepG2、THLE、HepaRG 细胞、原代人肝细胞），其终点从细胞毒性标志物（如 LDH、ATP）到使用高含量成像（HCI）和多种机制评估的方法不等。

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nature communications

Article

https://doi.org/10.1038/s41467-025-65690-3

A large-scale human toxicogenomics resource for drug-induced liver injury prediction

Received: 8 January 2025

Accepted: 21 October 2025

Published online: 13 November 2025

Check for updates

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Drug-Induced Liver Injury (DILI) remains one of the most critical challenges in drug development, causing patient safety concerns, clinical trial failures and drug withdrawals. We introduce *ToxPredictor*, a toxicogenomics framework combining RNA-seq data from primary human hepatocytes with pharmacokinetic data to predict dose-resolved DILI risks and safety margins. At its core is *DILImap*, an RNA-seq library tailored for DILI research, comprising 300 compounds at multiple concentrations. *ToxPredictor* achieves 88% sensitivity at 100% specificity in blind validation, outperforming state-of-the-art methods. It flagged recent phase III clinical failures, including Evobrutinib, TAK-875, and BMS-986142, overlooked by animal studies. Beyond prediction, *ToxPredictor* provides mechanistic insights into hepatotoxic pathways, enabling early de-risking and actionable safety decisions. Unlike single-endpoint readouts—even from 3D models—transcriptomics offers a multi-dimensional system-level view of hepatocyte responses, capable of detecting diverse DILI mechanisms not captured by conventional assays. Scalable, actionable, and integrated into a broader AI/ML drug discovery platform, this work establishes toxicogenomics as a promising tool for developing safer therapeutics and addressing one of the most pressing challenges in toxicology.

Drug-Induced Liver Injury (DILI) presents a poorly understood late-stage challenge in drug development, costing an estimated $350 million annually per pharmaceutical company[1]. Its rarity and unpredictability in clinical populations, often less than 1 in 10,000 persons, hinder detection in clinical studies, masking its severity until post-market exposure. Animal models fail to identify about half of the pharmaceuticals that exhibit clinical DILI[2]. This makes DILI a leading cause of drug candidate failure and market withdrawal, impeding the development of new therapies[3].

DILI arises from complex, multifactorial mechanisms, involving dose-dependent intrinsic mechanisms and with current methods unpredictable idiosyncratic reactions[4,5] influenced by genetic predisposition, environment, and individual health status[6]. DILI involves various cellular disruptions including mitochondrial dysfunction, oxidative stress, bile acid imbalance, inhibition of specific enzymes or transporters, and reactive metabolites formation[7,8]. However, the precise mechanisms and contributing factors are not fully delineated[4] and the lack of biomarkers hampers early detection[9]. Preclinical methods, such as quantitative structure-activity relationship (QSAR) models, offer low specificity and binary predictions lacking mechanistic insights[10,11]. In vitro models use diverse cell sources (e.g., HepG2, THLE, HepaRG cells, primary human hepatocytes) with endpoints ranging from cytotoxicity markers (e.g., LDH, ATP) to mechanistic assessments using high-content imaging (HCI) and multi-

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https://doi.org/10.1038/s41467-025-65690-3

参数化策略。3D肝脏模型旨在更好地模拟人体组织特征和体内反应[12]。然而，尽管生理相关性有所提升，这些模型仍然受制于低维度的读出——通常只是有限的标记物组合，例如ATP水平、LDH释放或基于成像的特征——无法捕捉分子反应的全谱，且常常无法发现机制原因。这一空白仍然导致药物在后期被撤回和临床失败，凸显了对更全面和可预测的DILI模型的迫切需求[9]。

受到将细胞视为复杂系统的理念启发，我们采用机器学习驱动的毒理基因组学方法来分析与DILI相关的化合物如何改变基因表达，从而识别出指示肝损伤的早期基因特征。涵盖毒性化合物响应中通路的相互作用，我们旨在破译DILI背后的分子机制。利用诸如DILIrank[13]和LiverTox[14]等资源将药物分类为DILI阳性和阴性，我们使用TG-GATES[15]微阵列数据库进行了初步概念验证，使我们能够在盲法验证中以62%的敏感性和92%的特异性准确预测DILI。在这一初步成功的基础上，我们创建了DILImap，这是一个专门构建且显著扩展的转录组库，旨在捕获更广泛的DILI机制谱。DILImap包含来自300种化合物在不同浓度下对初级人肝细胞（PHHs）进行分析的全转录组RNA-seq数据，使其

ToxPredictor 是我们基于随机森林的机器学习模型，在 DILImap 上训练，在盲法验证中实现了 88% 的敏感性（29/33 DILI 阳性）和 100% 的特异性（14/14 DILI 阴性）。在与 20+ 预临床模型的正面比较中，它表现优于包括机制性试验 [16–21]、细胞毒性标志物 [22–27]、理化性质 [28]、生物活化 [29]、BSEP [30] 方法以及最新的体外计算模型 [31–33]，能有效识别传统模型此前忽略的药物 DILI 风险。据我们所知，它是第一个在高关注度临床失败病例中提示 DILI 风险的预临床模型，例如 Evobrutinib、TAK-875 和 BMS-986142，这些药物尽管在预临床阶段表现良好，但最近在 III 期临床试验中因肝损伤被撤回。该模型提供剂量分辨的预测和机制洞察，展示了其在优先选择更安全药物候选物方面的实用性。

除了对未见化合物的泛化能力外，该模型在其机制范围上具有明显优势。该模型利用完整的转录组景观来检测与DILI相关的机制，例如线粒体功能障碍、氧化应激、免疫激活和代谢紊乱——通常在细胞死亡之前就已出现。这些优势在与高内涵三维肝脏检测[26,34–36]相比时尤为明显，尽管三维检测在生理上具有相关性，但通常仅限于低维的存活率或成像终点。在直接对比中，我们的模型能够独特地识别三维检测未能发现的非细胞毒性风险。这种系统水平的分辨能力使得在多种化合物类别中能够进行更全面、无偏的毒性风险检测。

这项工作是更广泛的人工智能/机器学习药物发现平台的重要组成部分，旨在提高药物开发中的预测能力和操作效率。它展示了从单一靶点向系统级视角的转变具有巨大潜力，并将机器学习在毒理基因组学中的应用定位为对现有方法的重大增强。为了推动该领域的发展并促进协作创新，我们已在 dilimap.org 上公开了我们的开源模型和验证数据，为降低药物候选物风险提供了强有力的工具，并为安全性评估的范式转变奠定基础。

结果

DILImap——用于DILI建模的人类毒理基因组数据库 我们创建了DILImap，这是一个为药物性肝损伤（DILI）建模量身定制的综合RNA-seq库，涵盖了以四种浓度测试的300种化合物。作为迄今为止最广泛的毒理基因组资源，DILImap包括经过精心挑选的DILI阳性和DILI阴性化合物，这些化合物涵盖了广泛已知的DILI机制，包括有充分文献记录的肝损伤药物以及没有特征性标记的特异性化合物（图1A）。

所有化合物均在三明治培养的初代人肝细胞（PHHs）中进行筛选，PHHs是评价肝毒性的金标准和最具生理相关性的体外模型，能够保留关键肝功能，如代谢活性和胆小管形成[37,38]。每种化合物在六个浓度下进行三次重复测试，使用乳酸脱氢酶（LDH）和三磷酸腺苷（ATP）细胞活力测定。RNA-seq分析在四个选定剂量下进行，剂量范围涵盖药理学相关范围，从治疗性血浆Cmax到刚低于IC10阈值的最高耐受非细胞毒性剂量（补充图S1）。

我们选择了暴露后24小时作为观察时间点，这是基于信号强度与细胞活力的权衡：较早的时间点（例如2小时或8小时）会产生较弱的转录反应[39]，而

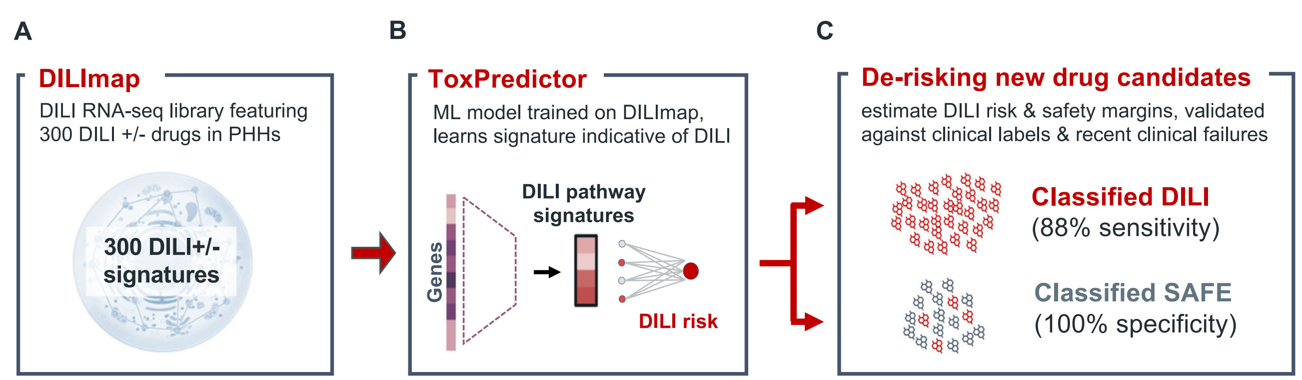

图 1 | DILImap 能够准确预测药物性肝损伤 (DILI)。A 我们创建了 DILImap，这是迄今为止最大的毒理基因组学数据集，涵盖了在初级人肝细胞中以多种浓度测试的 300 种化合物的 RNA-seq 数据。化合物涵盖多种 DILI 机制、特异性效应以及非 DILI 对照，确保了广泛的机制覆盖。该资源构成了训练我们 DILI 模型的基础。B ToxPredictor，我们的基于DILImap训练的机器学习模型能够学习指示DILI风险的通路特征。CToxPredictor通过评估DILI风险和安全边际，为新药候选物降低风险。在盲法验证中，其灵敏度达到88%（检测到29/33例DILI），特异性达到100%（所有14种非DILI化合物被准确分类）。这一框架为安全性评估带来了范式转变的可能。

https://doi.org/10.1038/s41467-025-65690-3

parametric strategies. 3D liver models aim to better mimic human tissue characteristics and in vivo responses[12]. However, despite improved physiological relevance, these models remain constrained by low-dimensional readouts–typically a limited panel of markers such as ATP levels, LDH release, or imaging-based features–which fail to capture the full spectrum of molecular responses and often miss the mechanistic cause. This gap continues to result in late-stage drug withdrawals and clinical failures, highlighting the urgent need for more comprehensive and predictive DILI models[9].

Inspired by the idea of viewing cells as complex systems, we adopt a machine learning-driven toxicogenomics approach to analyze how DILI-associated compounds alter gene expression, identifying early gene signatures indicative of liver injury. Encompassing the interplay of pathways in response to toxic compounds, we aim to decipher the molecular mechanisms underlying DILI. Utilizing resources like DILIrank[13] and LiverTox[14] for drug categorization into DILI positives and negatives, we employed the TG-GATES[15] microarray database for an initial proof-of-concept, enabling us to accurately predict DILI with 62% sensitivity and 92% specificity in blind validation. Building on this initial success, we created *DILImap*, a purpose-built and significantly expanded transcriptomic library designed to capture a broader spectrum of DILI mechanisms. DILImap features full-transcriptome RNA-seq data from 300 compounds profiled at multiple concentrations in primary human hepatocytes (PHHs), making it, to the best of our knowledge, the largest toxicogenomics dataset available for DILI modeling.

ToxPredictor, our random forest-based machine learning model trained on *DILImap*, achieves 88% sensitivity (29/33 DILI positives) at 100% specificity (14/14 DILI negatives) in blind validation. It outperforms 20+ pre-clinical models in a head-to-head comparisons, including mechanistic assays[16–21], cytotoxicity markers[22–27], physicochemical properties[28], bioactivation[29], BSEP[30] approaches, and the latest in-silico models[31–33], effectively identifying DILI risks in drugs previously overlooked by traditional models. To the best of our knowledge, it is the first pre-clinical model to flag DILI risks in high-profile clinical failures, such as Evobrutinib, TAK-875 and BMS-986142, all recently withdrawn in phase III trials due to liver injury despite clean preclinical profiles. The model provides dose-resolved predictions and mechanistic insights, demonstrating its utility for prioritizing safer drug candidates.

Beyond generalization to unseen compounds, the model has a distinct edge in its mechanistic breadth. The model leverages the full transcriptomic landscape to detect DILI-related mechanisms such as mitochondrial dysfunction, oxidative stress, immune activation, and metabolic perturbation–often well before cell death. These advantages are particularly evident when compared to high-content 3D liver assays[26,34–36], which, while physiologically relevant, are typically constrained to low-dimensional viability or imaging endpoints. In head-to-head comparisons, our model uniquely identifies non-cytotoxic risks missed by 3D assays. This systems-level resolution enables more comprehensive and unbiased detection of toxic liabilities across diverse compound classes.

This work, integral to a broader AI/ML drug discovery platform, aims at enhancing predictive power and operational efficiency in drug development. It showcases that the shift from a single-target to a systems-level perspective holds great promise and positions machine learning in toxicogenomics as significant enhancement to existing methods. To advance the field and foster collaborative innovation, we have made our open-source model and validation data publicly available at dilimap.org, providing a powerful tool for de-risking drug candidates and setting the stage for a paradigm shift in safety evaluations.

Results

DILImap – a human toxicogenomics database for DILI modeling

We have created *DILImap*, a comprehensive RNA-seq library tailored for drug-induced liver injury (DILI) modeling, encompassing 300 compounds tested at four concentrations. As the most extensive toxicogenomics resource to date, DILImap includes a curated selection of DILI-positive and DILI-negative compounds that span a wide range of known DILI mechanisms, including well-documented liver-injuring drugs and idiosyncratic compounds with no characteristic signature (Fig. 1A).

All compounds were screened in sandwich-cultured primary human hepatocytes (PHHs), the gold standard and most physiologically relevant in vitro model for liver toxicity, which preserve key hepatic functions such as metabolic activity and bile canaliculi formation[37,38]. Each compound was tested in triplicate across six concentrations using lactate dehydrogenase (LDH) and Adenosine Triphosphate (ATP) cell viability assays. RNA-seq profiling was performed at four selected doses, spanning the pharmacologically relevant range from therapeutic plasma Cmax to the highest tolerated non-cytotoxic dose just below the IC$_{10}$ threshold (Supplementary Fig. S1).

We selected a 24-hour post-exposure time point, based on the trade-off between signal strength and cellular viability: earlier time points (e.g., 2 h or 8 h) yield weaker transcriptional responses[39], while

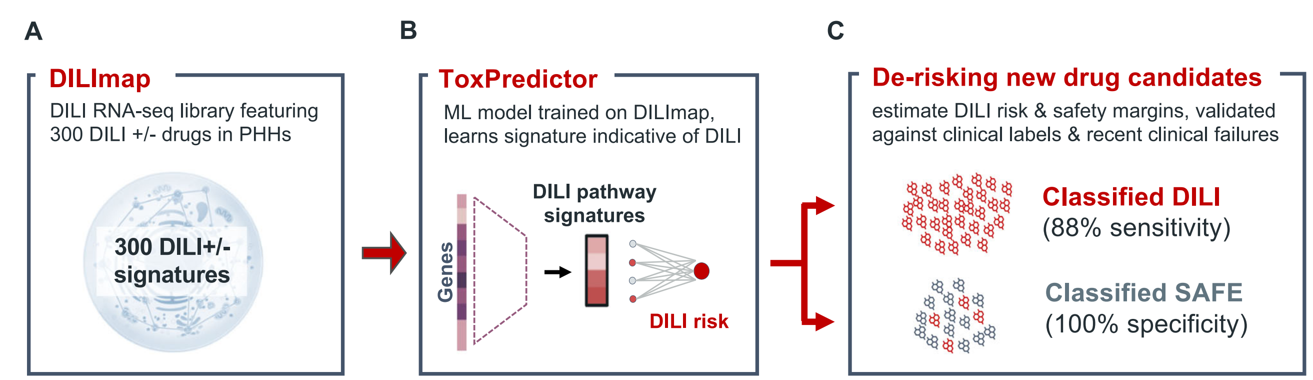

Fig. 1 | DILImap enables accurate prediction of drug-induced liver injury (DILI). **A** We created *DILImap*, the largest toxicogenomics dataset to date, encompassing RNA-seq profiles from 300 compounds tested at multiple concentrations in primary human hepatocytes. Compounds span a range of DILI mechanisms, idiosyncratic effects, and non-DILI controls, ensuring broad mechanistic coverage. This resource forms the foundation for training our DILI model. **B** *ToxPredictor*, our machine learning model trained on the DILImap, learns pathway signatures indicative of DILI risk. **C** ToxPredictor de-risks new drug candidates by estimating DILI risk and safety margins. In blind validation, it achieved 88% sensitivity (29/33 DILIs detected) and 100% specificity (all 14 non-DILI compounds accurately classified). This framework sets the stage for a paradigm shift in safety evaluations.

较长时间的孵育会增加肝细胞去分化和RNA降解的风险[40]。这种策略使我们能够在不损害RNA完整性的情况下捕获早期转录反应。标记分析确认在24小时内肝细胞特性得以保留。我们进一步通过仅包含总RNA计数足够且线粒体RNA含量低的孔来确保数据质量，这表明细胞是可存活且转录活跃的。这一简化工作流程——包括自动溶解性测试、活性筛选以及基于IC10的剂量选择——使我们能够在四个月内对300种化合物进行分析，其中包括110种首次在系统性基准测试中进行临床前试验的药物（补充图S2；方法）。为了支持全面的基准测试，我们提供了每种化合物的详细注释，包括临床DILI标签[13]、、DILI机制[14]、分子信息[41]、、来自各种研究的共识血浆Cmax[10],19,20,[23-25],28,42,[43]以及DILI分类结果。

根据 DILIrank[13] 和 LiverTox[14]，化合物被分类如下（补充图 S3）：

- 撤回DILI：由于DILI而撤回或临床试验失败（最关注DILI；已撤回）。
- 已知药物性肝损伤（DILI）：具有明确临床DILI风险的化合物（高DILI关注或LiverTox评分A）。
- 可能的药物性肝损伤（DILI）：有记录肝损伤病例的药物（最关注DILI或LiverTox评分A/B）。
- 特发性药物性肝损伤：罕见病例，无明确剂量–反应关系（肝脏毒性评分C/D，<12 病例报告）。
- 不太可能的DILI：各数据库证据不一致或较弱（较少关注DILI，但LiverTox评分为E）。
- 无DILI：无已记录肝毒性的化合物（无DILI关注；LiverTox评分E）。

撤回、已知和可能造成药物性肝损伤（DILI）作为阳性对照；无DILI作为阴性对照；而特异性和不太可能造成DILI的，由于其标签模糊，被排除在训练之外，并保留用于下游测试。

训练数据集包括249种化合物（111种DILI+，52种DILI-，17种不太可能引起DILI，69种特异性DILI）用于交叉验证。对于盲法验证，使用一个独立的51种化合物的实验集（33种DILI+，14种DILI-，以及4种标签未知的化合物，包括现实世界中的临床失败案例）进行了单独实验。这一精心策划的数据集为DILI的预测建模和机制性洞察提供了坚实的基础。

ToxPredictor – 基于通路水平的毒理基因组学预测DILI风险和治疗安全边际

ToxPredictor 是一个在 ourDILImap 库上训练的机器学习模型，它可以根据通路水平的转录特征预测药物诱导肝损伤（DILI）风险。这些特征是通过富集分析（WikiPathways[44]，FDR 调整后的 p 值）获得的，分析对象为通过 DESeq2[45]，计算的在化合物处理组与 DMSO 处理组样本中差异表达的基因，针对 DILImap 中每种化合物的每个剂量进行计算（图 1B；见方法部分）。

为了模型训练，我们仅使用具有明确 DILI 标签的化合物，从而得到一个高置信度的训练集，包括 111 个 DILI+(已撤回、已知、可能) 和 52 个 DILI−(无 DILI)，其余训练数据则被保留用于评估模型的稳健性。为了确保 DILI 标签的高置信度，我们进一步限制训练使用的药物浓度为其临床 Cmax 的 20 倍以上，以降低在较低剂量下可能表现安全的 DILI+ 化合物被误标为阴性的风险。在 5 折交叉验证中，我们采用分层、化合物级别的拆分，以确保每个化合物的所有剂量和重复样本都在每一折中一起保留，从而模拟对未知化合物的真实泛化。从 8 个模型类别中测试的 193 个配置中，我们选择了随机森林分类器，因为它具有较高的验证 AUC、最小的过拟合以及跨折的最高一致性。这些特性，加上其可解释性促使了它相对于更复杂的提升和深度学习模型的选择（补充图 S4）。

最终模型是由30个随机森林模型（集成成员）组成的集成模型，这些模型分别在不同的交叉验证切分上训练，从而共同提高了泛化能力和预测稳定性。集成规模的选择基于经验性基准测试，结果显示测试AUC稳定且模型间一致性良好（补充图S5）。通过在不同剂量水平上估计DILI概率，ToxPredictor能够计算药物安全边际，定义为首次预测的DILI剂量（即预测概率≥0.7的最低剂量）与治疗水平下的最大血浆浓度（Cmax）之间的比值。这提供了基于转录组学的临床治疗窗的替代指标。安全边际阈值为80，可将化合物进行高风险和低风险的可操作性分类。概率阈值0.7和安全边际（MOS）截止值80均在保留的训练数据上进行了优化，以在性能达到平台期的同时最小化假阳性率。

所有模型选择、超参数调优和阈值优化均仅在训练数据上进行。对于最终评估，我们使用了一个完全独立的盲验证集，包括51种化合物（33种DILI +，14种DILI −，以及4种未知化合物），这些化合物在一次独立实验中使用不同的板和测序运行进行分析。该数据集在模型开发的所有阶段均未使用。验证研究的化合物选择在训练前已确定，并特意丰富了撤市药物和近期临床失败药物。四种未知化合物代表目前在临床使用或试验中且尚未确认具有DILI风险的化合物（补充表S1）。

在盲法验证中，该模型达到了88%的敏感性，正确识别了33种DILI+化合物中的29种，并且达到了100%的特异性，所有14种DILI− 化合物都被正确分类为安全（图1C）。

识别先前在动物和临床研究中被忽视的撤回性和特异性药物性肝损伤

这些结果比我们最初在TG-GATES微阵列数据上训练的概念验证模型有了显著改进，该模型的敏感性为62%，特异性为92%。利用我们的DILImap库，ToxPredictor在相同的验证集上显著提高了敏感性至88%和特异性至100%（图2A）。在整个库的交叉验证中，该模型识别出144个DILI中的110个——超过了之前TG-GATES的144个中62个——并且仅错误分类了66个非DILI中的8个（图2B，补充图S7）。这种提升归因于DILImap拥有更大的数据集、更广泛的机制覆盖范围，以及RNA测序相对于微阵列的更高分辨率，使得基因检测更好，并且相对于微阵列在表达水平变化上有更广的定量范围。值得注意的是，我们的模型最有信心地将那些在临床前模型和临床试验中被遗漏的上市后撤回药物标记为高DILI风险。

我们的DILI安全边际和分类是由三个参数得出的：Cmax（基线浓度）、细胞活力测定（指示细胞死亡）和转录组学（基于差异通路）。为了评估每个参数的贡献，我们评估了它们独立分类DILI病例的能力。在144例DILI中，有29例

longer incubations risk hepatocyte de-differentiation and RNA degradation[40]. This strategy allowed us to capture early transcriptional responses without compromising RNA integrity. Marker analysis confirmed retention of hepatocyte identity at 24 hours. We further ensured data quality by including only wells with sufficient total RNA counts and low mitochondrial RNA content, indicating viable, transcriptionally active cells. This streamlined workflow—including automated solubility testing, viability screening, and IC_{10}-based dose selection—enabled us to profile 300 compounds in four months, including 110 drugs tested preclinically for the first time as part of a systematic benchmark (Supplementary Fig. S2; Methods). To support comprehensive benchmarking, we provide detailed annotations for each compound, including clinical DILI labels[13], DILI mechanisms[14], molecular information[41], consensus plasma Cmax from various studies[10,19,20,23–25,28,42,43], and DILI classification results from over 20 preclinical studies (Supplementary Data S1–S4).

Compounds were categorized based on DILIrank[13] and LiverTox[14] as follows (Supplementary Fig. S3):

- *Withdrawn DILI*: withdrawals or clinical trial failures due to DILI (Most-DILI-Concern; withdrawn).
- *Known DILI*: compounds with well-established clinical DILI risk (Most-DILI-Concern or LiverTox score A).
- *Likely DILI*: drugs with documented liver injury cases (Most-DILI-Concern or LiverTox score A/B).
- *Idiosyncratic DILI*: rare cases without clear dose–response (LiverTox score C/D, <12 case reports).
- *Unlikely DILI*: discordant or weak evidence across databases (Less-DILI-concern, but LiverTox score E).
- *No DILI*: compounds with no documented hepatotoxicity (No-DILI-Concern; LiverTox score E).

Withdrawn, *Known*, and *Likely DILI* serve as positive controls; *No DILI* as negative controls; while *Idiosyncratic* and *Unlikely DILI*, due to their label ambiguity, are excluded from training and reserved for downstream testing.

The training dataset includes 249 compounds (111 DILI+, 52 DILI-, 17 unlikely DILI, 69 idiosyncratic DILI) for cross-validation. For blind validation, a separate experiment was conducted using an independent set of 51 compounds (33 DILI+, 14 DILI-, and 4 with unknown labels, including real-world clinical failures). This carefully curated dataset provides a robust foundation for predictive modeling and mechanistic insights into DILI.

ToxPredictor – pathway-level toxicogenomics predicts DILI risk and therapeutic safety margins

ToxPredictor, a machine learning model trained on our *DILImap* library, predicts DILI risk from pathway-level transcriptional signatures. These signatures are derived through enrichment analysis (WikiPathways[44], FDR-adjusted p-values) of genes differentially expressed between compound- vs. DMSO-treated samples using DESeq2[45], computed for each dose of every compound in DILImap (Fig. 1B; see Methods).

For model training, we used only compounds with unambiguous DILI labels, resulting in a high-confidence training set of 111 DILI+ (*Withdrawn*, *Known*, *Likely*) and 52 DILI− (*No DILI*), while the remaining training data were held out to assess model robustness. To ensure high-confidence DILI labels, we further restricted training to drug concentrations tested at more than 20x of their clinical Cmax to reduce the risk of false-negative labeling for DILI+ compounds that may appear safe at lower doses. For 5-fold cross-validation, we applied stratified, compound-level splitting to ensure that all doses and replicates of a given compound were held out together in each fold, mimicking real-world generalization to unseen compounds. From 193 tested configurations across eight model classes, we selected a Random Forest classifier for its strong validation AUC, minimal overfitting, and highest consistency across folds. These properties, combined with its interpretability, motivated its choice over more complex boosting and deep learning models (Supplementary Fig. S4).

The final model is an ensemble of 30 Random Forest models (ensemble members) trained on different cross-validation splits, which together enhance generalization and prediction stability. The ensemble size was chosen based on empirical benchmarking that showed stable test AUC and consistency between models (Supplementary Fig. S5). By estimating DILI probabilities across dose levels, ToxPredictor enables calculation of drug *safety margins*, defined as the ratio between the first predicted DILI dose (i.e., the lowest dose with predicted probability >0.7) and the maximum plasma concentration (Cmax) at therapeutic levels. This provides a transcriptomics-based surrogate of the clinical therapeutic window. A safety margin threshold of 80 provides an actionable classification into high- and low-risk compounds. The probability threshold of 0.7 and margin of safety (MOS) cutoff of 80 were both optimized on held-out training data to reach performance plateaus while minimizing false positives. Our selected MOS threshold of 80, while on the higher end of literature-reported ranges (10–100)[17,23,28,34,36], reflects the greater sensitivity of transcriptomic assays compared to cytotoxicity or mechanistic readouts. Since transcriptional changes often occur at lower doses—before overt toxicity—a higher cutoff is needed to avoid false positives and maintain high specificity in a transcriptome-based model (Supplementary Fig. S5). Among available exposure measures, we used total Cmax instead of free Cmax due to its broader availability across compounds. Both measures showed comparable predictive performance, with total Cmax performing slightly better, possibly due to more robust consensus values derived from a greater number of studies (Supplementary Fig. S6).

All model selection, hyperparameter tuning, and threshold optimization were performed exclusively on the training data. For final evaluation, we used a fully independent blind-validation set of 51 compounds (33 DILI+, 14 DILI−, and 4 unknowns), profiled in a separate experiment using separate plates and sequencing runs. This set was withheld from all stages of model development. Compound selection for the validation study was finalized prior to training and intentionally enriched for withdrawals and recent clinical failures. The four unknowns represent compounds currently in clinical use or trials without confirmed DILI liability (Supplementary Table S1).

In blind validation, the model achieved 88% sensitivity, correctly identifying 29 of the 33 DILI+ compounds, and 100% specificity, with all 14 DILI− compounds correctly classified as safe (Fig. 1C).

Identifying withdrawn and idiosyncratic DILIs previously overlooked in animal and clinical studies

These results represent a substantial improvement over our initial proof-of-concept model trained on TG-GATES microarray data, which achieved 62% sensitivity and 92% specificity. Leveraging our DILImap library, ToxPredictor substantially improved both sensitivity of 88% and specificity of 100% on the same validation set (Fig. 2A). In cross-validation of the entire library, the model identified 110 out of 144 DILIs—surpassing the previous 62 out of 144 with TG-GATES—and misclassified only 8 out of 66 non-DILIs (Fig. 2B, Suppl. Figure S7). This enhancement is attributed to *DILImap's* larger dataset with broader mechanistic coverage and the higher resolution of RNA-seq over microarrays, enabling better gene detection and a wider quantitative range for expression level changes compared to microarrays[46]. Notably, post-market withdrawals, missed in both pre-clinical models and clinical trials, were most confidently flagged by our model as high DILI risk.

Our DILI safety margin and classification is derived from three parameters: *Cmax* (baseline concentration), *cell viability assays* (indicating cell death), and *transcriptomics* (based on differential pathways). To assess each parameter's contribution, we assessed their ability to classify DILI cases independently. Out of 144 DILIs, 29 were

https://doi.org/10.1038/s41467-025-65690-3

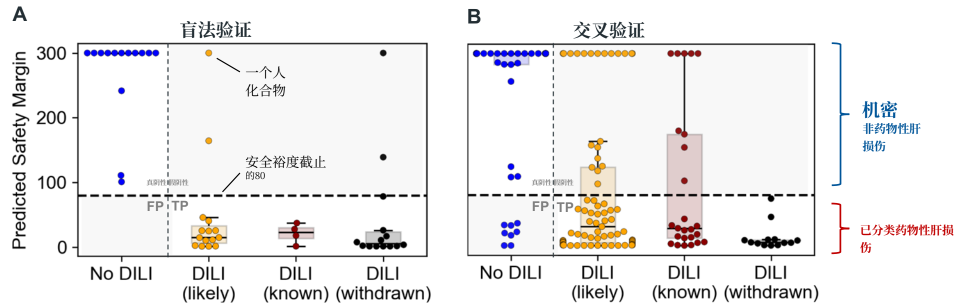

图 2 | 转录组学获得的安全边际能够准确识别药物性肝损伤（DILI）风险，包括已撤回的化合物。A 化合物根据DILI潜力分为四类：无DILI（n =14）、可能DILI（n =15）、已知DILI（n =4）和因DILI撤回（n =14）。每个数据点代表单个化合物的安全边际。箱线图显示中位数（中心线）、四分位距（箱体）以及在1.5 × 四分位距内的数据（胡须）。安全边际——计算方法为首次毒性剂量与治疗最高血药浓度（Cmax）的比率——在使用80的安全边际阈值进行盲法验证时，能够有效区分DILI阳性化合物和非DILI化合物(虚线)。该模型在识别DILI阳性和非DILI化合物方面显示出88%的敏感性和100%的特异性。B 对整个DILImap库进行交叉验证证实了模型在整个数据集上的稳健性能（无DILI: n = 52; 可能DILI: n = 73; 已知DILI: n = 25; 撤回: n = 13）。这些结果凸显了基于转录组学的建模在DILI检测方面超越传统方法的能力，为撤回和已知高风险化合物提供了更高的敏感性，同时保持对非DILI对照的高特异性。

仅基于血浆Cmax（>25 μM）和通过LDH细胞毒性测定检测出的42，特异性均为 ≥90%（安全边际 <80）。将我们的基于转录组的模型与Cmax和LDH数据结合使用最为有效，识别出了142例DILI中的110例（安全边际 <80），强调了毒理基因组学在DILI检测中的附加价值，超越了单纯的细胞死亡（补充图S8）。ToxPredictor在交叉验证中实现了0.82的ROCAUC，而单独使用细胞活力为0.66。在盲法验证中，其ROCAUC达到0.96，而仅使用细胞活力为0.65（补充图S9）。

对具有共同靶点但不同肝损伤（DILI）特征的非甾体抗炎药的机械性解析

我们的模型强调了密切相关的 COX-2 抑制剂非甾体抗炎药 (NSAIDs) 之间不同的 DILI 特征，并提供了将预测与机制（如肝细胞损伤、氧化应激和线粒体功能障碍）联系起来的独特机制性见解。例如，用于癌症疼痛的瓦德考昔布（Valdecoxib）显示没有 DILI 风险（图 3A），而用于关节炎的舒林酸（Sulindac）虽罕见但已确认为特异性 DILI 病例，以及因严重肝功能衰竭而被撤市的卢米考昔布（Lumiracoxib），则被标记为 DILI 风险，其安全边际低于分类阈值 80（图 3B）。

关键是，它强调了与药物性肝损伤（DILI）相关的途径，包括导致细胞损伤的直接因素如氧化应激，以及诸如脂肪酸代谢紊乱等间接因素，这对于解释特异性效应尤为相关（补充表 S2）。例如，舒林酸（Sulindac）与脂肪酸合成和胆固醇生物合成的紊乱相关，这与近期将其与肝脂肪变性相关联的研究一致[47]。通过明确这些途径，该模型为特异性 DILI 提供了机制性见解，从而提供了此前被认为无法预测的理解（图 3C）。

该模型以剂量分辨的方式评估药物性肝损伤（DILI）风险，揭示剂量如何影响肝损伤的可能性。一项关键示例是其对舒林酸（Sulindac）DILI风险的准确预测，尽管人们认为其DILI风险以剂量分辨方式不可预测[48]。这些结果突显了该模型提供可操作预测的能力，并通过聚焦关键通路（图3D）实现药物安全性特征的针对性优化。

与DILI风险相关的已知和新型基因及通路 DILI源于多种通路的紊乱。我们的模型突出了与DILI风险密切相关且预测价值高（AUC ≈ 0.8）的通路，包括氨基酸代谢（有毒代谢物堆积引起氧化应激和肝损伤）、脂肪酸生物合成（干扰导致脂质积累和肝细胞损伤）、色氨酸代谢（有毒中间产物引发氧化应激和炎症）以及铁死亡（依赖铁的氧化应激导致脂质过氧化物积累）[49–53]。此外，与预测的药物性肝损伤（DILI）风险高度相关的通路激活包括核受体信号通路（例如PXR/CAR/FXR）、一碳代谢以及胆汁酸调控——凸显转录重编程和代谢应激是肝毒性的关键因素[54–56] (图 4A)。当按预测DILI概率对化合物进行排序时，显示出通路激活的明显梯度，揭示这些生物过程的不同富集模式。这种相关性强化了模型预测的直接机制相关性和可解释性，并突出了这些通路作为潜在DILI驱动因素的作用（图 4B）。

为了识别对药物性肝损伤（DILI）最重要的基因，我们确定了在库中各DILI药物中每个基因上调或下调的频率，使用调整后的p值阈值为0.05。本分析集中于首次预测到毒性效应的浓度，旨在发现可能驱动DILI的早期上游调控因子。最常上调的基因与药物代谢、转运、应激反应和脂质代谢相关。还发现了一些与炎症、自噬和线粒体功能障碍相关的新基因（图4C）。频繁下调的基因包括对肝功能至关重要的基因，如药物代谢、转运、脂质代谢、氨基酸代谢、线粒体功能、凝血和炎症反应。这些基因表达的改变可能作为肝损伤的早期指标，并反映DILI的多方面机制（图4D）。

确定我们模型所识别的DILI通路和基因特异于肝脏毒性而非一般毒性，本质上是具有挑战性的。然而，该模型的精准性在于其能够准确分类具有已知其他系统毒性的非DILI化合物，例如瓦尔德昔布（心血管毒性）、布普洛尼（神经和心血管毒性）以及华法林（血液毒性）[57–59]，这表明模型能够区分肝脏特异性毒性与其他形式的器官损伤。

ToxPredictor 准确地标记了近期临床失败中的肝损伤风险，并提供剂量建议

布鲁顿酪氨酸激酶（BTK）抑制剂，尽管在肿瘤学和自身免疫性疾病中具有潜力，但由于肝损伤而面临临床暂停。最近的例子包括Evobrutinib、BMS-986142，以及

https://doi.org/10.1038/s41467-025-65690-3

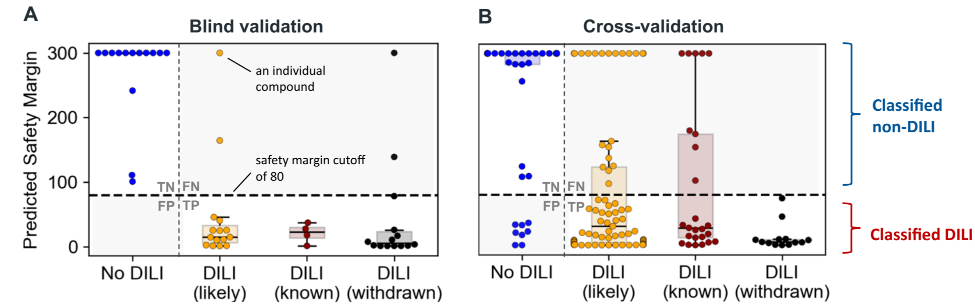

Fig. 2 | Transcriptomics-derived safety margins accurately identify DILI risk, including withdrawn compounds. A Compounds are grouped into four categories based on DILI potential: No DILI (n = 14), likely DILI (n = 15), known DILI (n = 4), and withdrawn due to DILI (n = 14). Each data point represents the safety margin of an individual compound. Boxplots show the median (center line), interquartile range (box), and data within 1.5 × interquartile range (whiskers). The safety margin—calculated as the ratio of the first toxic dose to the therapeutic maximum plasma concentration (Cmax)—effectively separates DILI-positive from non-DILI compounds in blind validation using a safety margin threshold of 80 (dashed line). The model demonstrated 88% sensitivity and 100% specificity in identifying DILI-positive and non-DILI compounds, respectively. **B** Cross-validation of the entire DILImap library confirms the model's robust performance across the entire dataset (No DILI: n = 52; likely DILI: n = 73; known DILI: n = 25; withdrawn: n = 13). These results highlight the ability of transcriptomics-based modeling to improve DILI detection beyond traditional methods, offering enhanced sensitivity for withdrawn and known high-risk compounds while maintaining high specificity for non-DILI controls.

detected solely based on plasma Cmax (>25 μM) and 42 through the LDH cytotoxicity assay (safety margin <80), both at ≥90% specificity. Combining our transcriptomics-based model with Cmax and LDH data was most effective, identifying 110 out of 142 DILIs (safety margin <80), underscoring the added value of toxicogenomics in DILI detection beyond mere cell death (Supplementary Fig. S8). ToxPredictor achieved a ROCAUC of 0.82 in cross-validation, compared to 0.66 for viability alone. In blind validation, it achieved a ROCAUC of 0.96, compared to 0.65 for using viability alone (Supplementary Fig. S9).

Mechanistic dissection of NSAIDS with shared targets but different DILI profiles

Our model highlights distinct DILI profiles among closely related COX-2 inhibitor non-steroidal anti-inflammatory drugs (NSAIDs) and imparts unique mechanistic insights linking predictions to mechanisms such as hepatocellular injury, oxidative stress, and mitochondrial dysfunction. For instance, Valdecoxib, used for cancer pain, shows no DILI risk (Fig. 3A), while Sulindac, an arthritis treatment with rare but established idiosyncratic DILI cases, and Lumiracoxib, withdrawn due to severe liver failures, are flagged as DILI risks with safety margins below the classification threshold of 80 (Fig. 3B).

Crucially, it highlights the pathways implicated in DILI, encompassing direct contributors like oxidative stress leading to cell injury, as well as indirect factors such as disturbances in fatty acid metabolism, which can be particularly relevant to explain idiosyncratic effects (Suppl. Table S2). Sulindac, for example, is linked to disruptions in fatty acid synthesis and cholesterol biosynthesis, aligning with recent studies connecting it to hepatic steatosis[47]. By pinpointing these pathways, the model provides mechanistic insights into idiosyncratic DILI, offering an understanding previously thought unpredictable (Fig. 3C).

The model assesses DILI risks in a dose-resolved manner, revealing how dosage impacts liver injury likelihood. A key demonstration is its accurate prediction of Sulindac's DILI risk, despite it being believed to be unpredictable in a dose-resolved manner[48]. These results highlight the model's ability to deliver actionable predictions and enable targeted optimization of drug safety profiles by focusing on critical pathways (Fig. 3D).

Known and novel genes and pathways associated with DILI risk

DILI arises from disruptions in diverse pathways. Our model highlights pathways with high predictive value (AUC ≈ 0.8) strongly associated with DILI risk, including *amino acid metabolism* (toxic metabolite buildup causing oxidative stress and liver injury), *fatty acid biosynthesis* (disruptions leading to lipid accumulation and hepatocyte damage), *tryptophan metabolism* (toxic intermediates driving oxidative stress and inflammation) and *ferroptosis* (iron-dependent oxidative stress leading to lipid peroxide accumulation)[49–53]. Additionally, pathway activations highly correlated with predicted DILI risk include *nuclear receptor signaling* (e.g., PXR/CAR/FXR), *one-carbon metabolism*, and *bile acid regulation*—highlighting transcriptional reprogramming and metabolic stress as key contributors to hepatotoxicity[54–56] (Fig. 4A). When compounds are ranked by predicted DILI probabilities, a clear gradient of pathway activation emerges, revealing distinct enrichment patterns for these biological processes. This correlation reinforces the direct mechanistic relevance and interpretability of the model's predictions and highlights these pathways as potential drivers of DILI (Fig. 4B).

To identify genes most significant for DILI, we determined the frequency at which each gene was differentially up- or downregulated across DILI drugs in our library, using an adjusted p-value threshold of 0.05. This analysis focused on the concentrations at which toxic effects were first predicted, aiming to uncover early upstream regulators potentially driving DILI. Most frequently up-regulated genes were associated with drug metabolism, transport, stress response, and lipid metabolism. Novel genes linked to inflammation, autophagy, and mitochondrial dysfunction were also implicated (Fig. 4C). Frequently down-regulated genes include those critical for liver functions such as drug metabolism, transport, lipid metabolism, amino acid metabolism, mitochondrial function, coagulation and inflammatory responses. Altered expression of these genes may serve as early indicators of liver injury and reflect DILI's multifaceted mechanisms (Fig. 4D).

Establishing that the DILI pathways and genes identified by our model are specific to liver toxicity rather than general toxicity is inherently challenging. However, the model's precision is evident in its accurate classification of non-DILI compounds with known toxicities in other systems, such as Valdecoxib (cardiovascular toxicity), Bupropion (neurologic and cardiovascular toxicity), and Warfarin (hematologic toxicity)[57–59], indicating the model's ability to distinguish liver-specific toxicity from other forms of organ damage.

ToxPredictor accurately flags DILI risks in recent clinical failures and provides dose recommendations

Bruton tyrosine kinase (BTK) inhibitors, despite their promise in oncology and autoimmune diseases, have faced clinical holds due to liver injury. Recent examples include Evobrutinib, BMS-986142, and

https://doi.org/10.1038/s41467-025-65690-3

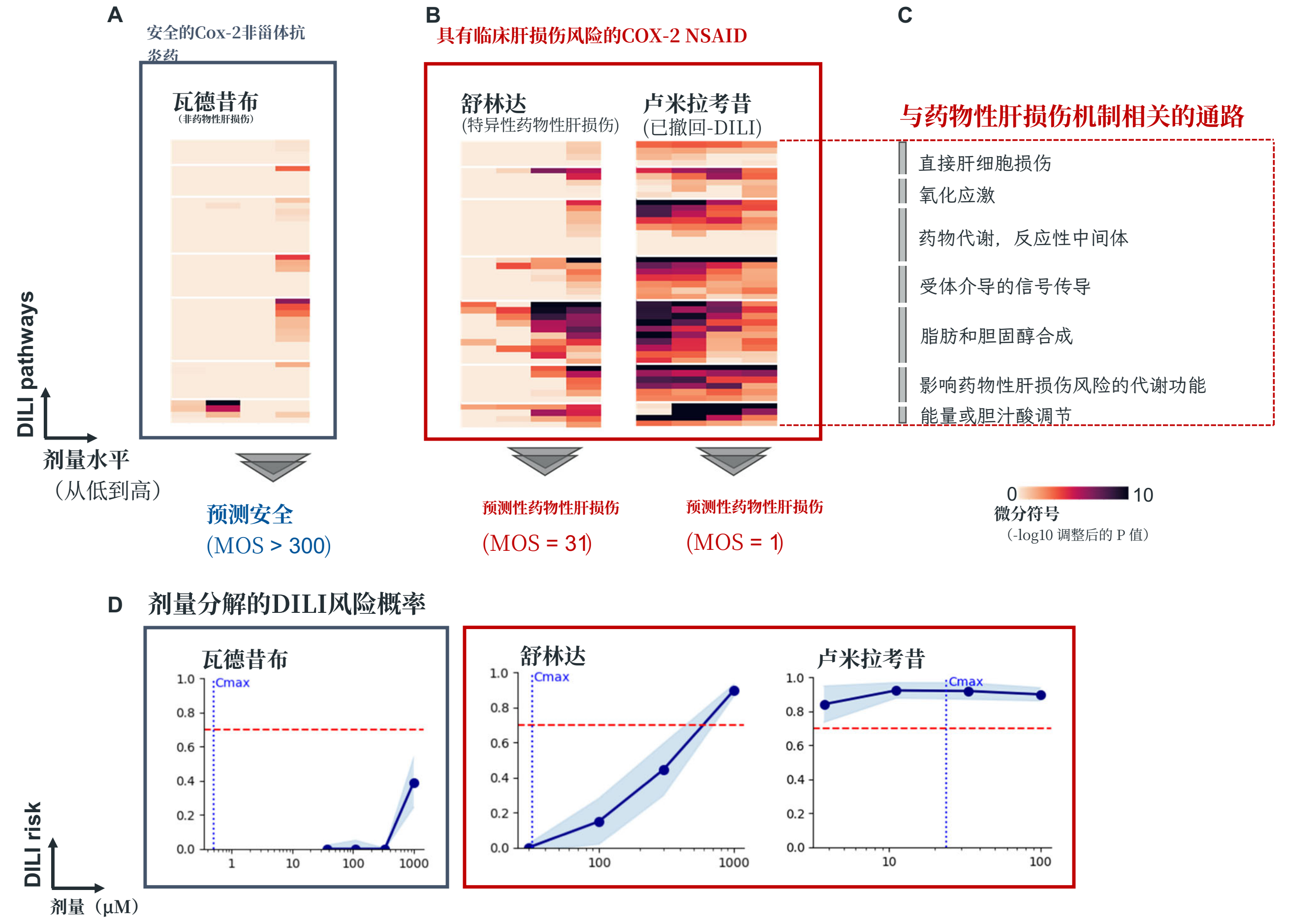

图 3 | 三种具有相同靶点但不同DILI（药物性肝损伤）特征的Cox2抑制剂。A 瓦地昔布（Valdecoxib），被认为安全，预测在所有剂量水平下没有DILI风险，安全边际（MOS）大于300。B 吲哚美辛（Sulindac），与特异性DILI相关，以及因严重肝衰竭而从市场上撤回的卢米洛昔布（Lumiracoxib），模型均标记为高风险，MOS值分别为31和1。C 预测结果由涉及DILI的通路驱动。差异性通路特征突出了DILI风险的关键机制，包括氧化应激、药物代谢、受体介导的信号传导、脂肪和胆固醇合成，以及胆汁酸调控。通路富集分析使用 Enrichr (gseapy) 进行，基于带有 Benjamini–Hochberg 校正的 Fisher 精确检验。颜色强度表示 −log10 校正后 p 值。D 剂量分辨的 DILI 风险概率展示了剂量增加如何影响肝损伤的可能性。点表示平均预测值，蓝色阴影置信区间表示集成模型的标准差。这些预测与临床结果高度一致，显示其在 DILI 风险分层、基于通路的机制理解以及指导更安全药物开发中的作用。

Orelabrutinib，所有这些在2023年由于DILI病例在III期被撤回或搁置。

我们在四个临床失败药物上验证了ToxPredictor：Evobrutinib、BMS-986142、Orelabrutinib和TAK-875（2型糖尿病药物），以及两个在研BTK抑制剂（Rilzabrutinib、Remibrutinib）和两个FDA批准的JAK抑制剂（Tofacitinib、Upadacitinib）作为阴性对照。在四种浓度下评估DILI风险概率，并估算安全边际（MOS）以将化合物分类为高风险（MOS ≤ 2 > 5）、中高风险（MOS ≤12 > 5）、中等风险（MOS ≤ 80）或低风险（MOS > 80）。所有临床失败药物均被标记为高风险或中高风险，MOS值较低，尤其是TAK-875、Evobrutinib和BMS-986142，这与其三期临床退出一致。在研药物被分类为中等风险（MOS = 14）和低风险（MOS = 101），目前尚未在临床研究中与DILI相关，而DILI阴性的JAK抑制剂被分类为低风险。这些结果与它们的临床安全特征高度一致（图5A）。

ToxPredictor 提供了基于不同假设 Cmax 值的经验性 DILI 可能性得出的剂量依赖性 DILI 风险曲线，从而能够提供安全剂量建议（见方法）。例如，Rilzabrutinib 在每日 100 mg 以下剂量被归类为低风险，这低于其有效剂量 400 mg。相比之下，Remibrutinib 的有效剂量 100mg 落在建议的低风险范围 <155mg q.d. 之内。这些发现突显了 ToxPredictor 在指导安全剂量策略方面的价值，使其成为降低新药候选风险的有价值工具（图 5B）。

在现有临床前DILI模型的竞争格局中绘制毒理基因组学图谱

我们沿着两个关键轴对我们的模型进行基准测试：预测性能和可扩展性。预测性能通过平衡准确率来衡量，反映模型区分DILI阳性和DILI阴性化合物的能力。可扩展性则包括技术吞吐量和生物学广度——即跨多样化化学物质和机制（包括先前未表征的机制）进行泛化的能力（图6A）。

我们的模型优于各种前临床DILI模型，包括机制性测定[16–21]、细胞毒性标志物[22–27]、理化特性[28]、生物活化[29] 和BSEP[30] 方法。在对匹配化合物集进行的正面比较中，它识别了66例DILI中的46例，而Xu等人识别了66例中的27例[17] HCI测定（49/66 对 27/66）；它在性能上优于Garside等人[20] HCI测定（46例中的37例对46例中的29例DILIs），Vorrink等人[26] 使用CD的细胞毒性测定

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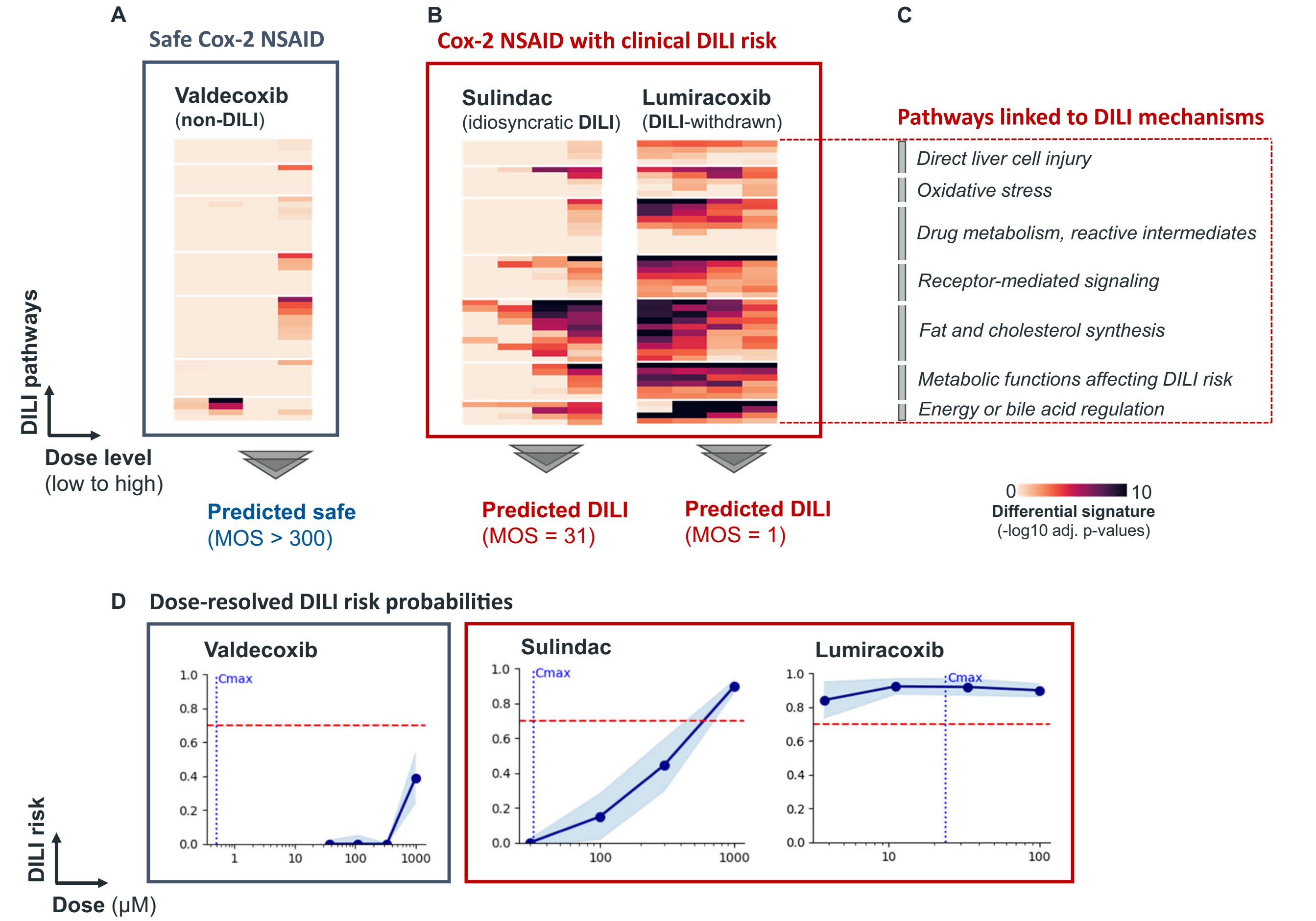

Fig. 3 | Three Cox2-inhibitors with the same target but distinct DILI profiles. **A** Valdecoxib, deemed safe, is predicted no DILI risk across all dose levels, with a margin of safety (MOS) greater than 300. **B** Sulindac, associated with idiosyncratic DILI, and Lumiracoxib, withdrawn from the market due to severe liver failure, are both flagged as high-risk by the model with MOS values of 31 and 1, respectively. **C** The predictions are driven by pathways implicated in DILI. Differential pathway signatures highlight key mechanisms underlying DILI risk, include oxidative stress, drug metabolism, receptor-mediated signaling, fat and cholesterol synthesis, and bile acid regulation. Pathway enrichment was performed using Enrichr (gseapy), based on a Fisher's exact test with Benjamini–Hochberg correction. Color intensity reflects −log10 adjusted *p*-values. **D** Dose-resolved DILI risk probabilities illustrate how increasing dosages impact the likelihood of liver injury. Dots indicate mean predictions and blue shaded confidence intervals represent the standard deviations across ensemble models. These predictions align closely with clinical outcomes, demonstrating its utility in DILI risk stratification, pathway-based mechanistic understanding, and guiding safer drug development.

Orelabrutinib, all of which were withdrawn or put on hold in phase III in 2023 due to DILI cases.

We validated *ToxPredictor* on four clinical failures: Evobrutinib, BMS-986142, Orelabrutinib, and TAK-875 (type 2 diabetes drug), along with two investigational BTK inhibitors (Rilzabrutinib, Remibrutinib) and two FDA-approved JAK inhibitors (Tofacitinib, Upadacitinib) as negative controls. DILI risk probabilities were assessed at four concentrations and margins of safety (MOS) estimated to classify compounds as high (MOS ≤ 2.5), mid-high (MOS ≤ 12.5), medium (MOS ≤ 80), or low risk (MOS > 80). All clinical failures were flagged as high or medium-high risk with low MOS values, particularly TAK-875, Evobrutinib, and BMS-986142, consistent with their phase III withdrawals. The investigational drugs were classified as medium risk (MOS = 14) and low risk (MOS = 101), which have not yet been linked to DILI in clinical studies yet, while the DILI-negative JAK inhibitors were classified as low risk. These results align closely with their clinical safety profiles (Fig. 5A).

ToxPredictor provides dose-dependent DILI risk curves derived from empirical DILI likelihoods across various hypothetical Cmax values, enabling safe dosing recommendations (see Methods). For instance, Rilzabrutinib is categorized as low risk at doses below 100 mg q.d., which is lower than its efficacious dose of 400 mg. In contrast, Remibrutinib's efficacious dose of 100 mg falls within the recommended low-risk range of <155 mg q.d. These findings highlight ToxPredictor's value in informing safe dosing strategies, making it a valuable tool for de-risking new drug candidates (Fig. 5B).

Mapping toxicogenomics in the competitive landscape of existing pre-clinical DILI models

We benchmark our model along two key axes: *predictive performance* and *scalability*. Predictive performance, measured by balanced accuracy, reflects the model's ability to distinguish DILI-positive from DILI-negative compounds. Scalability captures both technical throughput and biological breadth—the capacity to generalize across diverse chemistries and mechanisms, including previously uncharacterized ones (Fig. 6A).

Our model outperforms a wide range of pre-clinical DILI models, including mechanistic assays[16–21], cytotoxicity markers[22–27], physicochemical properties[28], bioactivation[29] and BSEP[30] approaches. In a head-to-head comparison across matched compound sets, it identified 46 out of 66 DILI cases versus the 27 out of 66 identified by Xu et al.[17] HCI assay (49/66 vs 27/66); it shows superior performance over Garside et al.[20] HCI assay (37/46 vs 29/46 DILIs), Vorrink et al.[26] cytotoxicity assay using CD

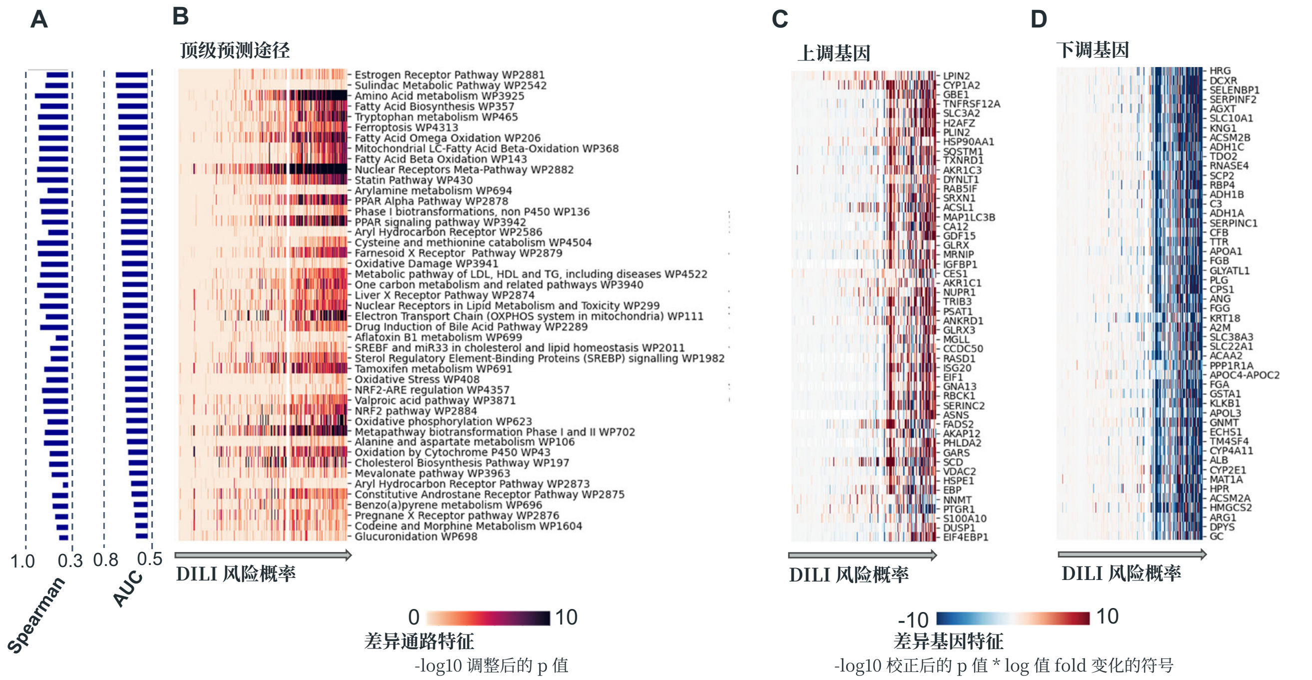

图 4 | DILI 中常见的关键通路激活和相关基因。A 通路根据其区分 DILI 与非 DILI 化合物的能力进行排名，衡量指标为 AUC。为了量化通路失调与预测 DILI 风险之间关联的强度，我们展示了化合物中通路激活评分与 DILI 概率之间的 Spearman 相关性。这突出显示了转录和代谢压力通路——如核受体信号、一碳代谢和胆汁酸调节——作为 DILI 风险的关键贡献因素。B 化合物按 DILI 风险概率排序，显示出 DILI 风险与与肝损伤相关的通路激活之间的相关性。通路富集分析使用 Enrichr（gseapy）进行，基于带 Benjamini–Hochberg 校正的 Fisher 精确检验。受影响最严重的通路包括对药物代谢、氧化应激和脂质稳态至关重要的通路。关键通路包括细胞色素 P450 氧化、NRF2。运输（例如，SLC3A2）、应激反应（例如，TXNRD1、SRXN1、GLRX、GLRX3、GCLM、HSP90AA1、HSP90AB1、HSPE1）和脂质代谢（例如，PLIN2、INSIG1、SREBF1、SCD、LPIN2、FADS2）。少数研究较少但重要的基因包括参与炎症（例如，GDF15、TNFRSF12A、S100A10、LITAF）、自噬体形成（例如，MAP1LC3B、SQSTM1）和线粒体功能（例如，VDAC2、RAN、CHCHD10）的基因。DILI病例中最常下调的前100个基因包括参与药物代谢和解毒（例如，CYP4A11、CYP2E1、AKR1C4、GSTA1、GSTA2、UGT2B10、UGT2B15）、药物转运（例如，SLC10A1、SLC22A1、SLC22A7、SLC38A3、SLC27A5）以及蛋白质处理（例如，ALB、AHSG、TTR、APOA1、APOE）的基因。与氨基酸代谢和线粒体功能相关的基因（例如，MAT1A、ARG1、CPS1、HMGCS2、ACAA1、ACAA2、ECHS1）也表现出显著下调。关键的氧化应激调节因子和氧化还原酶（例如，CAT、ABAT、GNMT 和 DHTKD1）也有所减少。
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类球体（37/43 对 30/43 DILIs），Sakatis 等人[29] 生物活化终点 GSH 加成物（47/65 对 25/65）以及他们结合共价结合和剂量的综合测定（47/65 对 32/65）。与 Kohonen 等人的基于转录组的细胞毒性模型[60]，相比，我们的方法显示出更高的敏感性（26/36 对 16/36 DILIs）。这些比较均在相同化合物上以 100% 特异性评估，突显了我们系统水平、机制不可知读出方法的附加价值（图 6B；补充表 S3）。

基于结构的计算机模拟模型，如 TxGemma[31]、DILIGeNN[32] 和 DILIPredictor[33] ，在我们的基准测试中表现不如体外方法。为了评估实际应用的泛化能力，我们在 314 种独立化合物（45 种 DILI +，269 种 DILI －）上对它们进行了评估，这些化合物主要通过 LiverTox 进行注释（评分 A/B 作为 DILI +，E 作为 DILI －）；TxGemma 还在扩展集合（143 种 DILI +，536 种 DILI －）上进行了测试。所有模型表现出有限的特异性：DILI-GeNN（84% 敏感性，28% 特异性）、DILIPredictor（80% 敏感性，29% 特异性）和 TxGemma-27B（57% 敏感性，37% 特异性）。这些发现略低于 Seal 等人（2024）报告的 DILIPredictor 平衡准确率 0.59。在与 DILI-map 重叠的未见化合物基准子集中（n = 97），TxGemma 的敏感性达到 63%（39/62），特异性为 57%（20/35），而我们的模型敏感性为 76%（47/62），特异性为 86%（30/35）。同样，DILIGeNN 在适中特异性（2/3）下显示出完美的敏感性（5/5），而我们的模型在两者上均达到 100%（5/5 和 3/3 ）。DILIPredictor 达到完全敏感性（ 23/23 ），但以较低的特异性（1/7）为代价，而 ToxPredictor 在明显更高的特异性（5/7）下保持高敏感性（20/23）。低特异性是基于结构的模型的一个主要限制，这类模型缺乏生物学背景，且倾向于高估毒性。这导致对常用且无肝毒性风险的药物出现假阳性，例如生物素（DILIPredictor 标记为 DILI+ ）、维生素 D（DILIGe 标记为 DILI+）

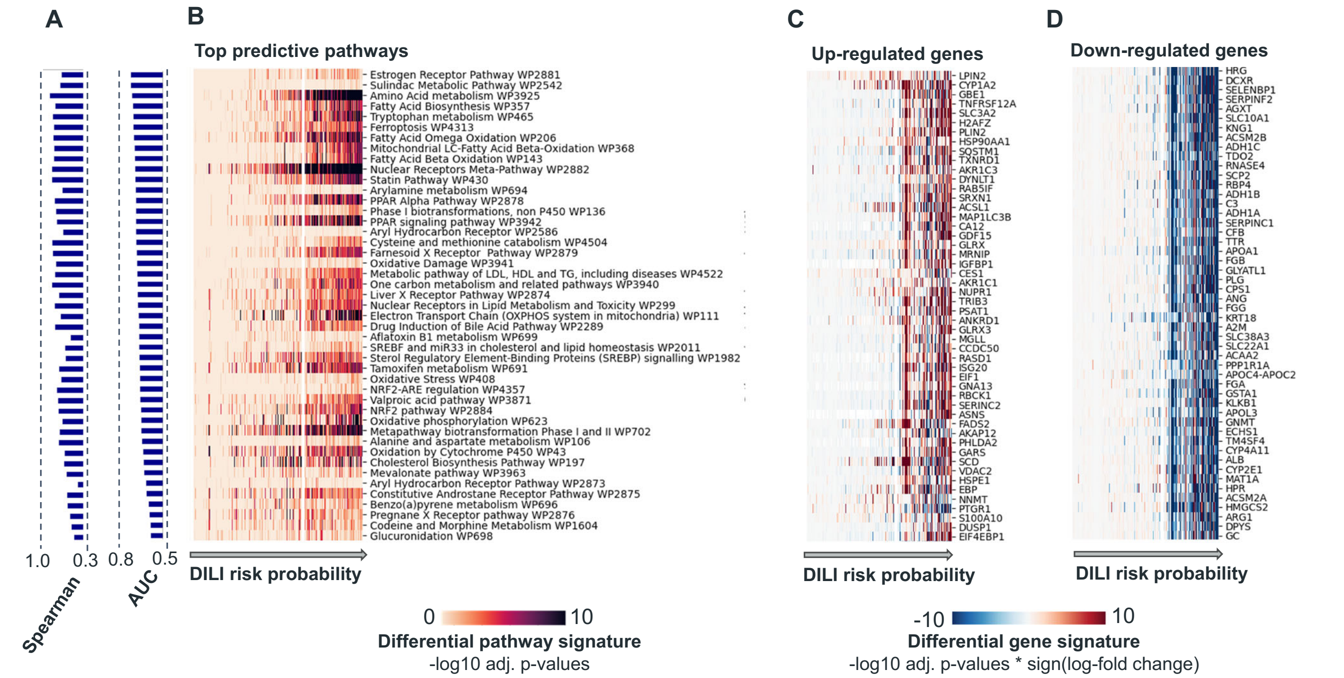

Fig. 4 | Key pathway activations and genes frequently implicated in DILI. **A** Pathways are ranked by their ability to distinguish DILI vs. non-DILI compounds, measured by AUC. To quantify the strength of association between pathway dysregulation and predicted DILI risk, we show the Spearman correlation between pathway activation scores and DILI probability across compounds. This highlights transcriptional and metabolic stress pathways—such as nuclear receptor signaling, one-carbon metabolism, and bile acid regulation—as key contributors to DILI risk. **B** Compounds are ordered by their DILI risk probability, showing a correlation between DILI risk and the activation of pathways associated with liver injury. Pathway enrichment was performed using Enrichr (gseapy), based on a Fisher's exact test with Benjamini–Hochberg correction. The most affected pathways include those critical to drug metabolism, oxidative stress and lipid homeostasis. Key pathways include Cytochrome P450 Oxidation, NRF2-ARE regulation for oxidative stress response, and Fatty Acid Beta-Oxidation, highlighting the interplay between detoxification, mitochondrial function, and energy metabolism. Additionally, nuclear receptor pathways such as Pregnane X Receptor (PXR) and Sterol Regulatory Element-Binding Protein (SREBP) signaling reveal disruptions in lipid and bile acid regulation. **C** Differential gene signatures were computed with DESeq2 using the Wald test, with Benjamini–Hochberg correction. Values are shown as −log10 adjusted p-values multiplied by the sign of the log-fold change. Top 100 genes most frequently upregulated in DILI cases include those related to drug metabolism (e.g., *CYP1A2, CYP51A1, UGT1A8, AKR1C1, AKR1C2, AKR1C3*), drug transport (e.g., *SLC3A2*), stress response (e.g., *TXNRD1, SRXN1, GLRX, GLRX3, GCLM, HSP90AA1, HSP90AB1, HSPE1*), and lipid metabolism (e.g., *PLIN2, INSIG1, SREBF1, SCD, LPIN2, FADS2*). Less commonly studied but significant genes include those involved in inflammation (e.g., *GDF15, TNFRSF12A, S100A10, LITAF*), autophagosome formation (e.g., *MAP1LC3B, SQSTM1*), and mitochondrial function (e.g., *VDAC2, RAN, CHCHD10*). **D** Top 100 genes most frequently downregulated in DILI cases include those involved in drug metabolism and detoxification (e.g., *CYP4A11, CYP2E1, AKR1C4, GSTA1, GSTA2, UGT2B10, UGT2B15*), drug transport (e.g., *SLC10A1, SLC22A1, SLC22A7, SLC38A3, SLC27A5*), and protein processing (e.g., *ALB, AHSG, TTR, APOA1, APOE*). Genes associated with amino acid metabolism and mitochondrial function (e.g., *MAT1A, ARG1, CPS1, HMGCS2, ACAA1, ACAA2, ECHS1*) also show significant downregulation. Key oxidative stress regulators and redox enzyme (e.g., *CAT, ABAT, GNMT, and DHTKD1*) are also reduced. Pathways related to lipid metabolism (e.g., *CIDEB, ACSM2A, ACSM2B, ACSM5*) and coagulation or inflammatory responses (e.g., *SERPINF2, SERPINC1, SERPINA6, SERPINA10, FGB, FGG, FGA*) are prominently affected. Notable genes with less established links to DILI but showing significant downregulation include *ITIH4, PGLYRP2, BHMT*, and *GUCA2B*, which could represent novel mechanisms or pathways contributing to the progression of liver injury. The changes in gene expression reflect early cellular alterations that may lead to DILI and highlight the complexity of DILI mechanisms, which encompass multiple aspects of liver function.

spheroids (37/43 vs 30/43 DILIs), Sakatis et al.[29] bioactivation endpoint GSH adduct (47/65 vs 25/65) as well as their combined assay integrating covalent binding and dose (47/65 vs 32/65). When compared to Kohonen et al.'s transcriptomics-based cytotoxicity model[60], our approach showed improved sensitivity (26/36 vs. 16/36 DILIs). These comparisons, all at 100% specificity evaluated on the same compounds, underscore the added value of our systems-level, mechanism-agnostic readout (Fig. 6B; Supplementary Table S3).

Structure-based in silico models such as TxGemma[31], DILIGeNN[32] and DILIPredictor[33] underperform in vitro-based approaches in our benchmark. To assess real-world generalizability, we evaluated them on 314 independent compounds (45 DILI + , 269 DILI −) primarily annotated via LiverTox (scores A/B as DILI + , E as DILI −); TxGemma was also tested on an expanded set (143 DILI + , 536 DILI −). All showed limited specificity: DILI-GeNN (84% sensitivity, 28% specificity), DILIPredictor (80% sensitivity, 29% specificity), and TxGemma-27B (57% sensitivity, 37% specificity). These findings are slightly below the balanced accuracy of 0.59 reported by Seal et al. (2024)[33] for DILIPredictor. On a benchmark subset of unseen compounds overlapping with DILI-map (n = 97), TxGemma reached 63% sensitivity (39/62) and 57% specificity (20/35), while our model achieved 76% sensitivity (47/62) and 86% specificity (30/35). Similarly, DILIGeNN showed perfect sensitivity (5/5) at moderate specificity (2/3), while our model reached 100% on both (5/5 and 3/3). DILIPredictor reached complete sensitivity (23/23) but at the expense of poor specificity (1/7), while ToxPredictor maintained high sensitivity (20/23) at markedly higher specificity (5/7). Low specificity is a key limitation of structure-based models, which lack biological context and tend to over-call toxicity. This results in false positives for commonly prescribed drugs with no risk of hepatotoxicity, such as biotin (flagged DILI+ by DILIPredictor), vitamin D (flagged DILI+ by DILIGeNN), and pemetrexed (flagged DILI+ by all three models). Moreover, they provide only binary outputs, without dose or mechanistic insight. In contrast, transcriptomics enables dose-

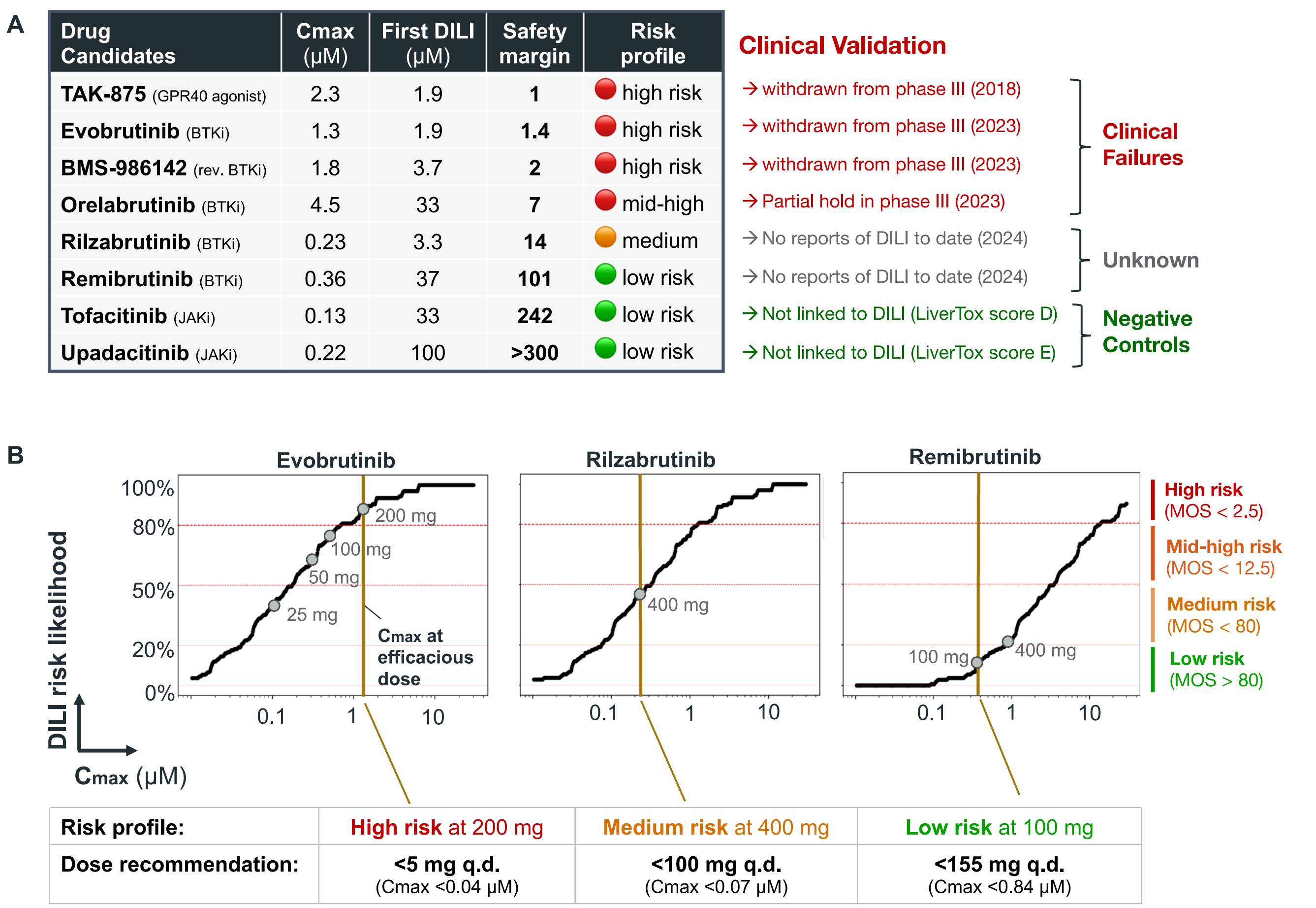

图 5 | 最近临床失败中DILI风险标记的现实应用。A. ToxPredictor提供的输出包括首次DILI浓度、安全边际以及相应的风险分类。临床验证确认，模型标记为高风险的药物，如TAK-875、Evobrutinib和BMS-986142，最近因DILI在III期试验中被撤回和中止。相反，模型分类为低风险的药物，如Tofacitinib和Upadacitinib，则没有DILI相关性。B. 代表性药物的模型推导DILI风险可能性曲线化合物相对于假设的Cmax值进行绘图。这些曲线描绘了与低、中、中高和高DILI风险相关的剂量方案，使在低风险范围内推荐剂量成为可能。风险分类是基于每种化合物的实际有效Cmax确定的：Evobrutinib被归类为高风险，Rilzabrutinib为中风险（推荐剂量 <100毫克每天一次），Remibrutinib为低风险（推荐剂量 <155毫克每天一次）。这展示了该模型在指导剂量选择以最小化DILI风险方面的实用性。

已解决的预测、机械可解释性和安全边际估计——对于评估毒性风险和指导后续实验至关重要（补充图 S10；补充数据 S4）。

3D肝脏系统提供了重要的生理学背景。在三维模型中的高内涵成像，例如Walker等人[34] 和Ewart等人[35], 的方法，在小型、精心挑选的测试组上表现出类似的性能（Walker: 23/27 对比 23/27；Ewart: 11/14 对比 13/14）。然而，它们的可扩展性有限，这限制了其在更广泛的药物性肝损伤机制中的应用。它们在狭窄、精选的测试组上可能表现良好，但在未知机制或低维终点未能覆盖的机制上可能表现不佳，如以下比较所示。为了探索二维转录组学与三维细胞毒性检测的独特能力，我们进行了与更大规模3D筛选研究的直接化合物水平对比：Vorrink等人[26] 和Fäs等人[36] 。在Vorrink等人的研究中，3D细胞毒性检测独特地发现了3种化合物（Fialuridine、Methotrexate、Trazodone），所有这些都与导致急性细胞死亡的细胞毒性效应相关。相反，我们的模型独特地标记了10种化合物——包括氟康唑、苯妥英钠以及化合物（例如西咪替丁、氟康唑、希美拉格特兰）表现出微妙的转录反应，而不是明显的细胞死亡。这些比较凸显了固定单终点模型的一个关键局限性：虽然在狭窄的情境下有效，但在更广泛的化学和机制多样性方面存在困难。相比之下，我们的转录组学方法提供了系统水平的分辨能力，能够适用于各类DILI通路——不仅可以检测已知的细胞毒性反应，还能发现传统检测常常遗漏的较不直接、非致死机制（补充数据S4）。

由于其无偏的建模方法，我们的方法在识别特异性化合物方面表现出了改进——这类毒性常常在针对性或表型狭窄的检测中被遗漏。这些化合物中，许多与极为罕见的临床发病率相关（<12 病例报道），为临床前筛查带来了重大挑战。我们的模型在65个此类病例中识别出了29个（44%），是所有评估模型中检测率最高的，同时保持了88%的特异性（补充图S11A）。

接下来，我们分析了将毒理基因组学与正交测定结合如何进一步增强检测能力。三种最有效的组合包括：将我们的模型与Walker等人的基于3D的HCI测定[34] 配对，将DILI检测从23/27例提高到26/27例；与Persson等人的基于2D的HCI测定[19] 配对，将DILI检测从28/37例提高到30/37例；以及与Sakatis等人的GSH耗竭测定[29] 配对，将检测数量从47/65提高到53/65。这样的

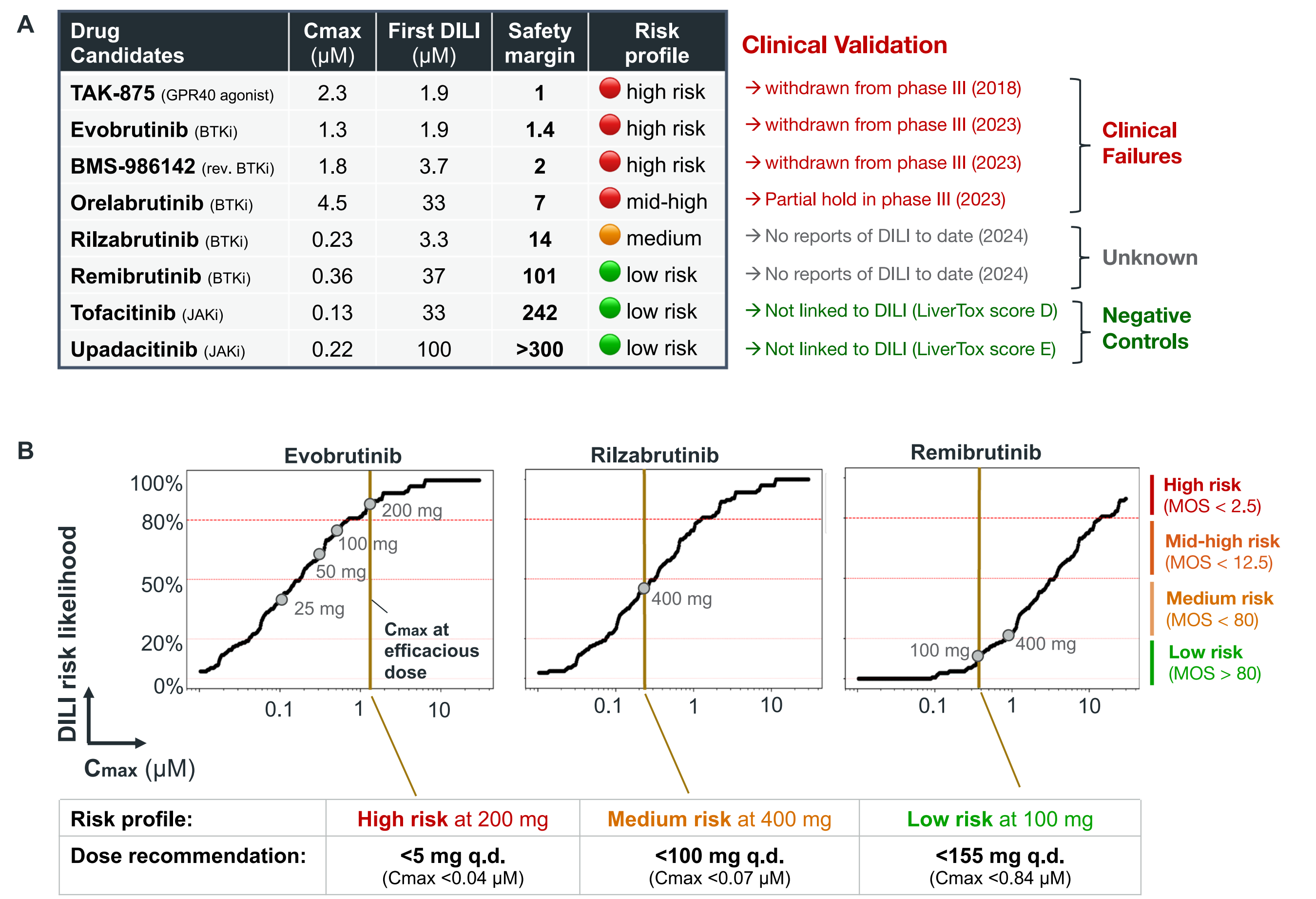

Fig. 5 | Real-world applicability in flagging DILI risk in recent clinical failures. A. ToxPredictor provides outputs including the first DILI concentration, safety margin, and corresponding risk classification. Clinical validation confirms that drugs labeled as high risk by the model, such as TAK-875, Evobrutinib, and BMS-986142, were recently withdrawn and halted in Phase III trials due to DILI. Conversely, low-risk drugs such as Tofacitinib and Upadacitinib, as classified by the model, exhibit no DILI association. **B** Model-derived DILI risk likelihood curves for representative compounds are plotted against hypothetical Cmax values. These curves delineate dose regimes associated with low, medium, mid-high and high DILI risk, enabling dose recommendations within the low-risk range. Risk classification is determined based on the actual efficacious Cmax of each compound: Evobrutinib is classified as high risk, Rilzabrutinib as medium risk (recommended dose <100 mg q.d.), and Remibrutinib as low risk (recommended dose <155 mg q.d.). This demonstrates the model's utility in guiding dose selection to minimize DILI risk.

resolved predictions, mechanistic interpretability, and safety margin estimation—critical for evaluating toxic liabilities and guiding follow-up experiments (Supplementary Fig. S10; Supplementary Data S4).

3D liver systems offer important physiological context. High-content imaging in 3D models, such as those by Walker et al.[34] and Ewart et al.[35], achieves similar performance on small, curated panels (Walker: 23/27 vs. 23/27; Ewart: 11/14 vs. 13/14). However, their limited scalability constrains their utility to a broader range of DILI mechanisms. They may perform well on narrow, curated panels, but struggle with unknown mechanisms or mechanisms not captured by the low-dimensional endpoint, as shown in the following comparison. To explore the unique capabilities of 2D transcriptomics vs 3D cytotoxicity assays, we conducted direct compound-level comparisons with larger 3D screening studies: Vorrink et al.[26] and Fäs et al.[36] In Vorrink et al. 3D cytotoxicity uniquely detected 3 compounds (Fialuridine, Methotrexate, Trazodone), all linked to cytotoxic effects that result in acute cell death. Conversely, our model uniquely flagged 10 compounds—including fluconazole, phenytoin, and zileuton—associated with immune activation, metabolic stress, or enzyme modulation, which are not readily captured by viability endpoints. A similar pattern emerged in the Fäs et al. study: 3D cytotoxicity exclusively identified 4 compounds (e.g., Haloperidol, Fialuridine) whose toxicities depend on structural or metabolic context. Our model uniquely identified 5 compounds (e.g., Cimetidine, Fluconazole, Ximelagatran) marked by subtle transcriptional responses rather than overt cell death. These comparisons highlight a key limitation of fixed single-endpoint models: while effective in narrow contexts, they struggle with broader chemical and mechanistic diversity. Our transcriptomic approach, by contrast, offers systems-level resolution that generalizes across DILI pathways—not only detecting known cytotoxic responses but also uncovering less immediate, non-lethal mechanisms often missed by traditional assays (Supplementary Data S4).

As a result of its unbiased modeling, our approach shows improved detection of idiosyncratic compounds—a class of toxicities that often escape detection in targeted or phenotypically narrow assays. These compounds, many of which are associated with extremely rare clinical incidence (<12 case reports), present a significant challenge for preclinical screening. Our model identified 29 out of 65 of those cases (44%) the highest detection rate among all evaluated models, while maintaining a specificity of 88% (Supplementary Fig. S11A).

Next, we analyzed how combining toxicogenomics with orthogonal assays further enhances detection. The three most effective combinations include pairing our model with Walker et al.'s 3D-based HCI assay[34] to improve DILI detection from 23/27 to 26/27 cases, pairing with Persson et al.'s 2D-based HCI assay[19] to improve DILI detection from 28/37 to 30/37 cases, and pairing with Sakatis et al. GSH depletion assay[29] to increase detection from 47/65 to 53/65. Such

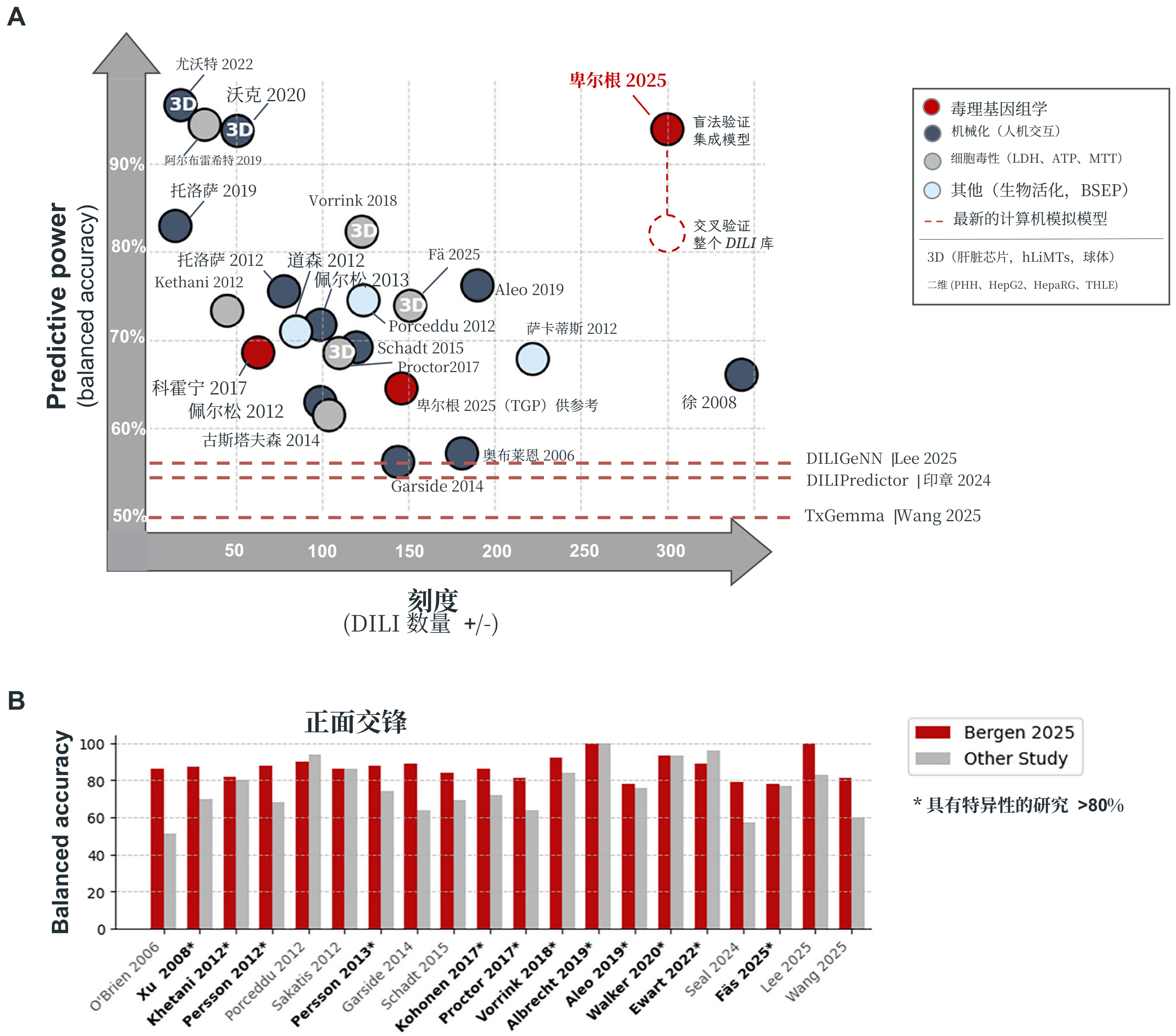

图 **6** | ToxPredictor 在准确性和可扩展性方面优于最先进的预测模型。A 平衡准确性与规模（已发布的临床前模型中 DILI +/− 化合物的数量）之间的关系。ToxPredictor 在体外和计算方法中均达到最高性能，优于高内涵成像、细胞毒性、生物活化及基于结构的模型，即使扩展到数百种化合物也表现出稳健的性能。B 在重叠化合物上的正面对比显示出相对于其他研究的一致性能提升。具有 >80% 特异性的模型（标记为 *）支持可信的临床前风险降低。结果强调了基于转录组学的毒理基因组学作为一种机制性知情、可扩展的 DILI 预测的领先策略。

战略组合可能将平衡精度提高到高达98%（补充图S11B）。

基于这些见解，我们提出了一种分层去风险漏斗策略，该策略从简单的终点指标开始，例如药代动力学数据（例如 Cmax <25 μM）和细胞毒性测定，以标记明显的肝毒性。对于未显示早期毒性信号的候选物，毒理基因组学提供了最全面且无偏见的 DILI 风险评估——捕捉已知和新颖的机制。对于少数高级候选物，如果有足够的资源，在高级 3D 肝脏模型中应用毒理基因组学可能提供对体内反应的最准确预测。这一策略确保了在药物开发中资源高效且机制广泛的 DILI 风险评估。

讨论

我们的毒理基因组学方法提供了对细胞反应的全面且无偏见的视角，为提供丰富的信息对肝脏毒性有细致入微的理解，包括机制不明的特异性反应。通过从单靶点分析转向系统级观点，我们展示了将机器学习应用于毒理基因组学具有巨大的潜力，能够显著提升现有毒理学方法。它能够进行剂量特异性的预测和安全边际评估，超越了单纯的DILI/非DILI的二元评估。其广泛的机制覆盖在现有的临床前模型中具有显著优势。通过创建专门用于DILI研究的综合毒理基因组学库，我们在预测DILI风险方面相比最先进的方法取得了显著改进。在盲法验证中获得的高敏感性（88%）和特异性（100%），以及在被撤市药物和动物及临床研究中被忽视的临床失败药物中识别DILI，强调了其在药物早期开发阶段的实际应用价值。我们认为

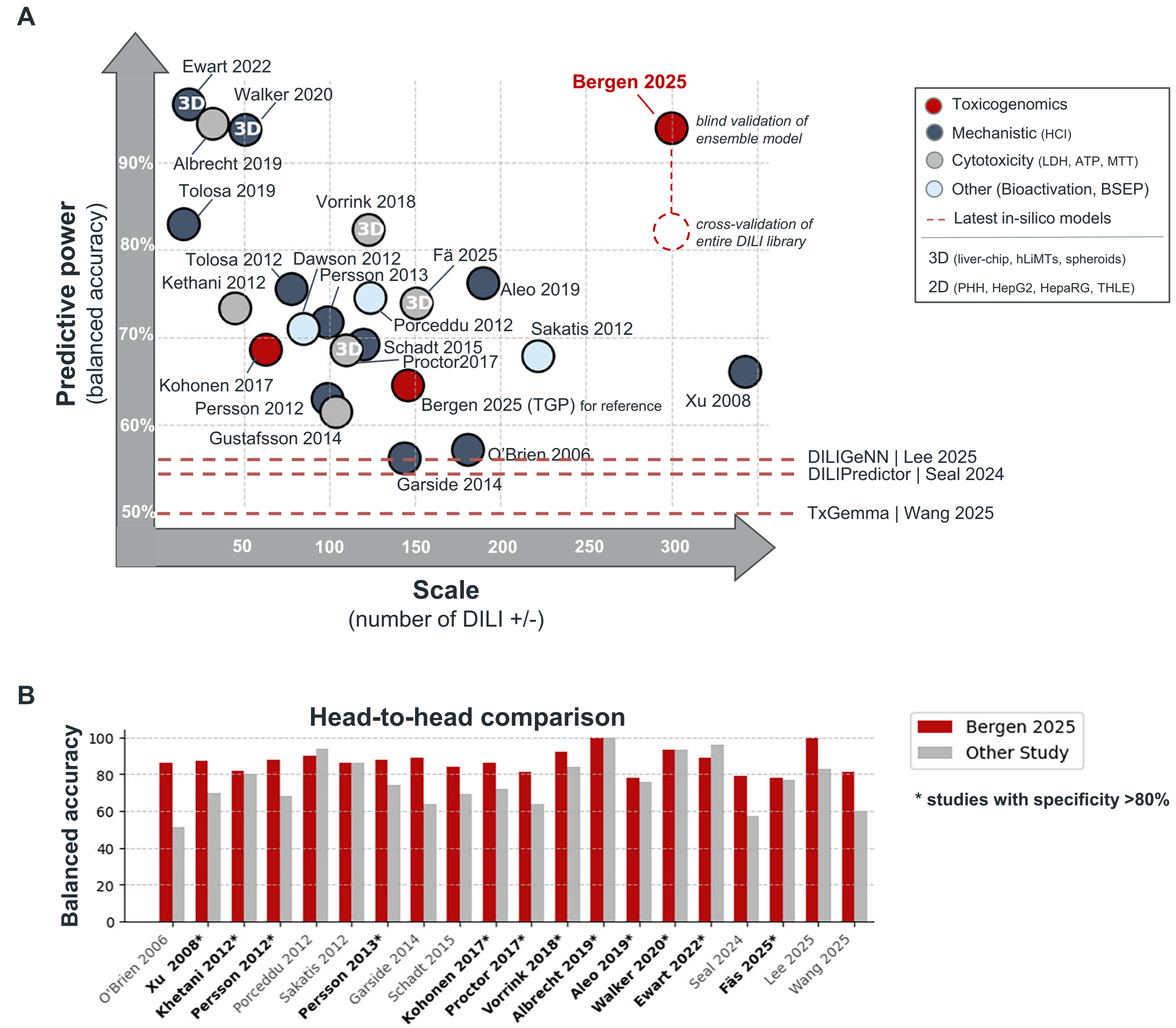

Fig. 6 | ToxPredictor outperforms state-of-the-art prediction models in accuracy and scalability. A Balanced accuracy versus scale (number of DILI + /− compounds) across published preclinical models. ToxPredictor achieves the highest performance among both in vitro and in silico methods, outperforming high-content imaging, cytotoxicity, bioactivation-based and structure-based models, demonstrating robust performance even when scaled to hundreds of compounds. **B** Head-to-head comparison on overlapping compounds shows consistent performance gains over other studies. Models with >80% specificity (marked with *) support credible preclinical de-risking. Results highlight transcriptomics-based toxicogenomics as a leading strategy for mechanistically informed, scalable DILI prediction.

strategic combinations could raise balanced accuracy to as high as 98% (Supplementary Fig. S11B).

Based on these insights, we propose a tiered de-risking funnel strategy that begins with straightforward endpoints, such as PK data (e.g., Cmax <25 μM) and cytotoxicity assays, to flag overt hepatotoxicity. For candidates showing no early toxicity signals, toxicogenomics provides the most comprehensive and unbiased assessment of DILI risk—capturing both known and novel mechanisms. For a select few advanced candidates, with sufficient resources, applying toxicogenomics in advanced 3D liver models may offer the most accurate prediction of in vivo responses. This strategy ensures a resource-efficient and mechanistically broad DILI risk assessment in drug development.

Discussion

Our toxicogenomics approach offers a comprehensive and unbiased perspective on cellular responses, providing rich information for a nuanced understanding of liver toxicity, including idiosyncratic reactions with unknown mechanisms. By shifting from single-target analyses to a systems-level viewpoint, we demonstrate that applying machine learning to toxicogenomics holds great promise as a significant enhancement over existing toxicology methods. It enables dose-specific predictions and safety margins, moving beyond binary DILI/No-DILI assessments. Its extensive mechanistic coverage presents a substantial advantage over current pre-clinical models. By creating a comprehensive toxicogenomics library specifically designed for DILI research, we achieved notable improvements in predicting DILI risks compared to state-of-the-art methods. The high sensitivity (88%) and specificity (100%) obtained in blind validation, along with the identification of DILI in withdrawn drugs and clinical failures overlooked in animal and clinical studies, underscore its practical utility in early drug development phases. We consider this an important step forward in predictive toxicology, positioning our approach as a significant advancement in the field. The scalability and adaptability of our

方法，作为更大人工智能/机器学习平台的核心组件，旨在增强药物开发流程的预测能力和效率。

虽然我们的方法代表了一个有意义的进步，但承认我们方法的固有限制仍然很重要。首先，DILI 的多因素特性涉及遗传、环境和生活方式因素，这意味着即使是最先进的模型也无法捕捉所有潜在机制的全谱。我们的方法虽然全面，但并未考虑所有个体间的差异或罕见的遗传易感性，这些都会影响 DILI 风险。其次，我们模型的预测能力受限于 DILImap 库的完整性。数据库中的缺口，尤其是与记录不充分的特异反应相关的部分，可能限制模型的准确性。第三，我们对二维肝细胞培养系统的依赖可能无法再现肝细胞与其他细胞类型或组织之间的复杂相互作用，而这种相互作用可能推动某些 DILI 机制的发生。第四，24小时的暴露时间点限制了对延迟或免疫介导毒性的检测。第五

展望未来，毒理基因组学在推进我们对药物性肝损伤（DILI）的理解和预测方面具有巨大潜力。将RNA测序与先进的三维培养系统相结合——例如球体和肝脏芯片平台——可能实现更长时间的药物暴露，并包括库普弗细胞和肝星状细胞等非实质细胞。这些共培养模型对于捕捉免疫激活、炎症和纤维化至关重要——这是肝细胞单一培养无法再现的特发性和慢性DILI的标志。由于长期三维模型依赖于高内涵成像或细胞毒性终点来捕捉特定表型反应，因此它们在机制范围和可扩展性方面仍然有限。它们的性能通常在小规模、有针对性的已知机制化合物集上得到验证，这可能无法推广到更广泛的化学领域。相比之下，转录组学提供了一种可扩展的、无偏的检测手段，能够发现多种DILI机制，包括那些三维模型未能捕捉到的机制。

将 RNA-seq 数据与分子结构信息相结合，有助于更深入地理解药物与细胞组分之间的相互作用，从而促进对毒性更准确的预测。使用早期估计值作为血浆 Cmax 的替代指标，例如靶点活性，可以在药物发现过程中更早地推导安全边际。开发包含 DILI 监管网络的多变量模型是另一个令人振奋的前沿。这类模型可以整合参与 DILI 的基因、蛋白质和代谢通路的复杂相互作用。此外，探索其他类型的数据，如染色质可及性、蛋白质组学或代谢组学，可能进一步深入理解 DILI 机制。

随着这些先进的模型和数据集被整合到药物开发的早期阶段，我们预计肝脏相关的不良事件将减少，药物开发效率将提高，成本将显著节约，更重要的是，患者安全性将得到增强。毒理基因组学方法的持续发展，借助计算机机器学习方法和多组学技术的进步，标志着向更具预测性的药物安全性评估迈出了重要的一步——这有可能支持更安全的治疗药物开发，并改善患者的临床结局。

方法

实验工作流程

我们使用冷冻保存的原代人肝细胞（PHHs）采用系统的高通量方法来研究药物性肝损伤（DILI）背后的转录变化。

人体组织采购与伦理合规

从 LifeNet Health（美国弗吉尼亚海滩）按照标准供应商协议获得冷冻保存的 PHHs。LifeNet Health 根据知情捐赠者同意和机构审查委员会（IRB）批准采购组织，符合美国法规。Cellarity 没有为本发布创建任何新的细胞系。

细胞培养概述

冷冻保存的人肝原代细胞（PHHs）被选择用于高活力（>95%）、长期贴壁性（10–15天）、与96孔板格式兼容以及A级质量。使用的供体为1917277-01，一名37岁的白人女性（体重指数26）。肝细胞在 I 型胶原涂层的96孔板中培养，并保持在夹心配置（胶原底层覆盖 100 μg/mL的Matrigel）以维持肝细胞功能。细胞在化合物处理前培养三天，每天更换培养基，使其成熟。处理24小时后，细胞或者用于活力检测（LDH/ATP），或者裂解用于RNA提取和测序（补充图S1A）。

细胞培养方案

人肝细胞解冻培养液、人肝细胞培养液、HHCM 补充液、人肝细胞铺板培养液、HHPM 补充液（LifeNet Health）以及青霉素/链霉素（Gibco）在铺板前解冻并过滤。使用 Luna-FL 细胞计数仪和吖啶橙/碘化丙啶染色（Logo Bio）对细胞进行计数。细胞以 0.5 百万细胞/毫升的密度铺板于 I 型胶原涂层的 96 孔板（Gibco）中。细胞在培养箱中孵育 6 小时以便附着，然后更换为维持培养液（LifeNet Health）。第二天，在细胞表面覆盖一层薄薄的 Matrigel，并继续孵育一天，每日更换培养液，以促进细胞完全成熟。第 3 天，用多种浓度的化合物处理细胞 24 小时，然后对细胞进行存活率检测（LDH 和 ATP 检测）或裂解以进行 RNA 测序。

实验装置

本实验的目的是筛选300种化合物，以建立一个用于构建预测性药物性肝损伤（DILI）模型的肝毒性干预库。首先，对原代人肝细胞（PHHs）进行了最大耐受剂量（MTD）筛选，使用六点对数曲线（0.01 μM至1 mM，图S1B）。MTD定义为在LDH检测中观察到 >10%细胞死亡之前的最高浓度。对处理过的细胞进行了RNA测序，化合物浓度范围从Cmax到MTD，用于评估治疗剂量与毒性剂量之间的安全边际。

化合物溶解和板材制备

所有化合物均购自 MedChemExpress (MCE)。化合物制备在内部完成。化合物溶解液在用药当天手工新鲜配制，以避免冻融循环。在之前的 DMSO 耐受性测试中，化合物中 0.5% 的 DMSO 被确定为理想浓度。使用 DMSO 溶解化合物时，初始浓度为最高 1000 μM。如果在 200 倍储备液下化合物未溶解，则尝试在 100 倍和 50 倍浓度下溶解。化合物被加入带有条形码的板中，并转移到 Hamilton MicroLab Star 液体处理器进行滴定（图 S1B）。滴定板布局在毒性筛选和 RNA 测序实验中有所不同（图 S1C）。自动化用于准备化合物稀释液和化合物处理。

method, a central component of a larger AI/ML platform, are designed to enhance the predictive power and efficiency of drug development pipelines.

While our approach represents a meaningful step forward, it is important to acknowledge the inherent limitations of our method. First, the multifactorial nature of DILI, involving genetic, environmental, and lifestyle factors, means even the most advanced models cannot capture the full spectrum of potential mechanisms. Our approach, though comprehensive, does not account for all inter-individual variability or rare genetic predispositions contributing to DILI risk. Second, the predictive power of our model is constrained by the completeness of the *DILImap* library. Gaps in the database, particularly in relation to poorly documented idiosyncratic reactions, can limit the model's accuracy. Third, our reliance on a 2D hepatocyte culture system may fail to replicate the complex interactions between hepatocytes and other cell types or tissues that can drive certain DILI mechanisms. Fourth, the exposure timepoint of 24 hours limit the detection of delayed or immune-mediated toxicities. Fifth, as with any preclinical model, the ultimate test of its utility lies in its ability to predict clinical outcomes, a domain where uncertainties and unpredicted variables can significantly impact performance.

Looking ahead, toxicogenomics holds tremendous promise for advancing our understanding and prediction of DILI. Integrating RNA-seq with advanced 3D culture systems—such as spheroids and liver-on-chip platforms—may enable longer drug exposures and the inclusion of non-parenchymal cells like Kupffer cells and hepatic stellate cells. These co-culture models are essential for capturing immune activation, inflammation, and fibrosis—hallmarks of idiosyncratic and chronic DILI that hepatocyte monocultures cannot recapitulate. As long 3D models rely on high-content imaging or cytotoxicity endpoints that capture specific phenotypic responses they remain limited in mechanistic scope and scalability. Their performance is often demonstrated on small, curated compound sets with known mechanisms, which may not translate to broader chemical space. In contrast, transcriptomics provides a scalable, unbiased readout capable of detecting diverse DILI mechanisms, including those not captured by existing assays. As RNA-seq becomes more feasible in physiologically relevant 3D systems, we anticipate a powerful synergy —combining the mechanistic breadth of transcriptomics with the physiological relevance of 3D models. Our work lays the foundation for such integration, demonstrating that transcriptomics alone can robustly capture DILI risk across a wide range of mechanisms.

Combining RNA-seq data with structural information of molecules could enable a deeper understanding of the interactions between drugs and cellular components, facilitating more accurate predictions of toxicity. Using early estimates as surrogate for plasma Cmax, such as target activity, could help derive safety margins earlier in the drug discovery process. The development of multivariate models that include DILI regulatory networks represents another exciting frontier. Such models can incorporate the complex interplay of genes, proteins, and metabolic pathways involved in DILI. Furthermore, exploring additional data types, such as chromatin accessibility, proteomics, or metabolomics, could yield further insights into DILI mechanisms.

As these advanced models and datasets become integrated into the early stages of drug development, we anticipate a decrease in liver-related adverse events, improved efficiency in drug development, significant cost savings, and, most importantly, enhanced patient safety. The ongoing evolution of toxicogenomics approaches, bolstered by advancements in computational machine learning methods and multi-omics technologies, marks an important step toward more predictive drug safety evaluation—one that has the potential to support the development of safer therapeutics and improved patient outcomes.

Methods

Experimental workflow

We employed a systematic, high-throughput approach using cryopreserved primary human hepatocytes (PHHs) to study transcriptional changes underlying drug-induced liver injury (DILI).

Human tissue sourcing and ethical compliance

Cryopreserved PHHs were obtained from LifeNet Health (Virginia Beach, VA, USA) under standard provider agreements. LifeNet Health procured tissues under informed donor consent and Institutional Review Board (IRB) approval in accordance with U.S. regulations. Cellarity did not create any new cell lines for this publication.

Cell culture overview

Cryopreserved PHHs were selected for high viability (>95%), long-term plateability (10–15 days), compatibility with 96-well formats, and Grade A quality. Donor 1917277-01, a 37-year-old Caucasian female (BMI 26), was used. Hepatocytes were cultured in collagen I-coated 96-well plates and maintained in a sandwich configuration (collagen base with a 100 μg/mL Matrigel overlay) to preserve hepatocyte function. Cells were matured for three days with daily media changes before compound treatment. After 24 h of treatment, cells were either assayed for viability (LDH/ATP) or lysed for RNA extraction and sequencing (Supplementary Fig. S1A).

Cell culture protocol

Human hepatocyte thawing media, human hepatocyte culture media, HHCM supplement, human hepatocyte plating media, HHPM supplement (LifeNet Health) and Pen/Strep (Gibco) were thawed and filtered before plating. PHHs were counted using a Luna-FL Cell Counter and Acridine Orange/Propidium Iodide Stain (Logo Bio). Cells were plated at a cell density of 0.5 million cells / mL in collagen I-coated 96-well plates (Gibco). Cells were left to attach in the incubator for 6 hours and then replaced with maintenance medium (LifeNet Health). The next day, cells were overlayed with a thin coat of Matrigel and left to incubate for an additional day with a daily media change to allow for full maturation of cells. On Day 3, cells were treated with compounds at multiple concentrations for 24 h and then were either taken down for either viability testing (LDH and ATP readouts) or lysed for RNA sequencing.

Experimental setup

The objective of this experiment was to screen 300 compounds to create a hepatotoxicity intervention library for building a predictive DILI model. PHHs were first screened for maximum tolerated dose (MTD) in six-point log curves (0.01 μM to 1 mM, Figure. S1B). MTD was defined as the highest concentration before observing >10% cell death in the LDH assay. RNA sequencing was conducted on cells treated with compound concentrations ranging from Cmax to MTD to evaluate the safety margin between therapeutic and toxic doses.

Compound dissolution and plate preparation

All compounds were purchased from MedChemExpress (MCE). Compound preparation was performed in-house. Compound dissolution was prepared manually fresh on the day of compound treatment to avoid freeze/thaw cycles. In a previous DMSO tolerance test, 0.5% of DMSO in compound was dictated as the ideal concentration. Compound dissolution with DMSO started at a highest concentration of 1000 μM. If compounds were not solubilized at the 200x stock, dissolution was attempted at 100x and 50x concentrations. The compounds were added to a barcoded plate and transferred to the Hamilton MicroLab Star liquid handler for titration (Fig. S1B). The plate layout for titration varied between toxicity screens and RNA sequencing runs (Fig. S1C). Automation was used to prepare compound dilution and compound treatment.

乳酸脱氢酶（LDH）活性检测
作为主要途径，我们使用了Promega的非放射性细胞毒性测定法，该方法利用乳酸脱氢酶（LDH）作为细胞死亡的标志。在试剂制备中，我们在4°C下将LDH缓冲液解冻过夜，并用其重构LDH底物瓶，加入12毫升，然后存储在−20°C。我们使用指定的未处理细胞建立100%裂解产物作为阳性对照，并以此标准化所有后续处理的细胞以计算存活率。向这些指定的细胞孔中，我们按1:10的比例加入10倍裂解缓冲液。然后我们将 50 μ微升收集的样品与 50 μ微升LDH底物混合在平底组织培养板中。培养板覆盖后，在室温下孵育30分钟。孵育结束后，我们加入 50 μ微升解冻的停止液，并使用SpectraMax i3x酶标仪，吸光度设置为490 nm，读取板子。

CellTiter-Glo (CTG) 活力测定
另一种用于毒性筛查的正交活力测定是CellTiter-Glo（CTG）发光细胞活力测定（Promega），其通过测定孔内的ATP含量来评估。CTG活力测定是终点测定，因为此过程会裂解细胞。配制试剂时，将CTG缓冲液和底物在室温下解冻，将10 mL缓冲液重新溶解于底物中，然后储存在-20 °C冰箱中。CTG等分试剂在实验当天解冻，并按1:1的比例向CTG底物中加入DPBS（Gibco）。在清除培养板内容物后，将100 μL CTG与PBS混合液加入板上的所有孔中。裂解后，使用铝箔覆盖培养板，并置于轨道摇床上以400 rpm摇动2分钟。2分钟后，培养板保持覆盖状态并在室温下放置10分钟。随后，使用SpectraMax i 3x在发光模式和标准不透明设置下读取培养板数据。

用于RNA提取的细胞裂解
准备 RLT 缓冲液（Qiagen）和 2-巯基乙醇（Thermofisher Scientific）以制备 RLT + 1% BME 试剂。使用多通道移液器将 140 μL 裂解缓冲液加入板子的所有孔中。在显微镜下确认细胞已裂解后，将 140 μL 裂解液转移到 Eppendorf twin.tec 96 孔 PCR 板（Fisher Scientific）中，并放入 −80°C 冰箱中。

文库制备与测序（SMART-Seq）
总RNA是使用QIAcube HT机器人工作站（Qiagen）结合RNeasy 96 QIAcube HT试剂盒（Qiagen）从96孔板中的细胞裂解液中制备的，操作遵循制造商推荐的方案。根据制造商的方案，从总RNA制备了SMART-Seq DE3文库。简而言之，对于每一行孔板样品，使用带有独特条形码的寡dT引物选择聚腺苷酸化的mRNA。通过逆转录生成第一链cDNA；通过模板切换结合有限循环的PCR扩增生成双链cDNA。然后将每行样品汇集，并使用Nextera XT DNA文库制备试剂盒（Illumina）进行转座酶介导的片段化。文库随后通过使用Illumina P5和P7条形码的独特组合进行PCR扩增，并在测序前以等摩尔池混合。测序在Illumina NovaSeq6000上进行，使用自定义的读取长度为89 bp（Read1）和26 bp（Read2）。

计算工作流程
Cmax 注释。Cmax 值是从多个资源汇编而来的，这些值的中位数被用来得出一致的总 Cmax。以下资源对计算一致的 Cmax 有贡献：来自国家转化科学推进中心（NCATS）的药物信息，Porceddu 等（2012），Khetani 等（2013），Persson 等（2013），Aleo 等（2014），Garside et al.（2014）、Gustafsson et al.（2014）、Chen et al.（2014）、Shah et al.（2015）、Camenisch et al.（2019）、Dixit et al.（2019）、Aleo et al.（2020）、Williams et al.（2020）和 Smit et al.（2020）。对于没有研究提供可靠 Cmax 估计值或各研究估计值差异显著的化合物，通过检索 PubChem 和相关临床研究进行额外的手动标注。

DILI 分类。DILIrank 和 LiverTox 是用于将药物分类为 DILI 阳性或阴性的重要资源。DILIrank 由 FDA 开发，根据历史肝损伤数据，将超过 1000 种药物分为四个 DILI 风险等级——高风险、较低风险、无风险和未确定。LiverTox 由 NIDDK 创建，是一个详尽的在线数据库，提供关于这些药物的详细信息，包括其临床表现、作用机制以及引起 DILI 的可能性。它还提供了一套评估 DILI 可能性的系统，从已知病例到无特征性 DILI 表现的个体化药物，再到被认为是安全的药物。这些分类作为我们模型开发和优化的 DILI 终点。化合物基于 DILIrank 和 LiverTox 被系统地分为以下几类：

因DILI撤回：由于DILI导致的市场撤回和临床失败（最关注DILI；例如，曲格列酮）。
- 已知DILI：具有明确DILI风险的化合物（在DILIrank中大多数为DILI关注或在LiverTox评分中为A；例如，治疗剂量的异烟肼，超量的对乙酰氨基酚）。
- 可能的药物性肝损伤（DILI）：在特定背景下有记录DILI病例的化合物（高度DILI关注或LiverTox评分A/B；例如，孕酮）。
- 特异性药物性肝损伤（DILI）：罕见、不可预测的 DILI，无明确的剂量反应（LiverTox 评分 C/D；<12 病例报告）。
- 不太可能的药物性肝损伤（DILI）：在 LiverTox 中被不一致地标记为安全（评分 E），但在 DILIrank 中被认为是较少关注 DILI 的药物。
- 无药物性肝损伤（DILI）：在临床或临床前没有药物性肝损伤的记录证据的化合物（DILIrank中无DILI关注；LiverTox评分E）。

类别“Withdrawn DILI”“Known DILI”和“Likely DILI”作为阳性对照，“No DILI”作为阴性对照。我们的模型经过训练以区分阳性和阴性对照化合物。由于标签不明确，类别“Unlikely DILI”和“Idiosyncratic DILI”被排除在模型训练之外，并保留用于后续测试。

在训练数据中，DILI 阳性包括最高浓度以及高于 20 倍 Cmax 的数据，而 DILI 阴性包括最低浓度以及低于 80 倍 Cmax 的数据。这种方法确保了标注的明确性，同时最大限度地减少了阴性中过量用药特征和阳性中非活性特征带来的偏差。使用这些分类，我们的模型被训练用于区分 177 个阳性和 93 个阴性对照数据点。另有一组 70 种已知存在特异性作用的化合物，每种化合物关联的病例报告少于 12 例，仅用于测试。特异性肝毒性的特征是不可预测，且缺乏剂量反应或时间模式，尽管经过充分的临床测试，仍常导致药物撤市。为了盲法验证，还进行了一项独立的内部实验，包括 46 种化合物：32 个 DILI 阳性和 14 个阴性。这包括公认的 DILI 阳性/阴性以及一个真实世界的数据集，其中有 4 个 re

毒理基因组学资源和初步努力。人类毒理基因组学受益于诸如 TG-GATES、CMap 和 L1000 等资源。CMap 和 L1000 记录了体外细胞系（主要是癌症细胞系）的基因组反应，可用于研究细胞毒性，然而，它们在药物性肝损伤（DILI）方面的效用有限，因为我们最初的研究显示预测性能较差，可能是由于缺乏生理相关的细胞类型和相关浓度范围。相比之下，TG-GATES 提供了一个专门针对 DILI 的丰富微阵列数据库。

Lactate dehydrogenase (LDH) viability assay

As primary viability assay, we employed the non-radioactive cytotoxicity assay from Promega, which utilizes lactate dehydrogenase (LDH) as a marker for cell death. For reagent preparation, we thawed LDH buffer at 4 °C overnight and used it to reconstitute the LDH substrate bottles by adding 12 mL, which were then stored at −20 °C. We utilized designated untreated cells to establish the 100% lysate positive control, against which we normalized all subsequently treated cells for % viability calculations. To these designated cell wells, we added 10x Lysis buffer in a 1:10 volume ratio. We then mixed 50 μL of the collected sample with 50 μL of LDH substrate in a flat-bottom tissue culture plate. Plates were covered by and incubated at room temperature for 30 min. Following the incubation, we added 50 μL of thawed Stop solution and used a SpectraMax i3x plate reader with absorbance settings at 490 nm to read the plate.

CellTiter-Glo (CTG) viability assay

Another orthogonal viability assay that was utilized for toxicity screening was CellTiter-Glo (CTG) Luminescent Cell Viability Assay (Promega) which determined the ATP content within the wells. The CTG viability assay is a terminal endpoint due to the lysis of cells. To prepare the reagent, the CTG buffer and substrate were thawed at room temperature and 10 mL of the buffer was resuspended in the substrate and stored at −20 °C freezer. CTG aliquots were thawed the day of the takedown, and DPBS (Gibco) was added to CTG substrate at a 1:1 ratio. Following the removal of the contents of the culture plate, 100 uL of CTG and PBS mix was added across all the wells of the plate. Following the lysis, plates were covered with aluminum foil and set on an orbital shaker for 2 min at 400 rpm. Following the 2 min, the plates remained covered and were left at room temperature for 10 minutes. The plates were then read on the SpectraMax i3x using the luminescence and Standard Opaque settings.

Cell lysis for RNA extraction

RLT buffer (Qiagen) and 2-mercaptoethanol (Thermofisher Scientific) were prepared to create a RLT + 1% BME reagent. 140 uL of lysis buffer was added across all wells of the plate using a multichannel pipette. After ensuring the cells were lysed under a microscope, the 140 uL of lysate was transferred to an Eppendorf twin.tec 96-well PCR plate (Fisher Scientific) and placed into a −80 °C freezer.

Library preparation and sequencing (SMART-Seq)

Total RNA was prepared from cell lysates in 96-well plates using a QIAcube HT robotic workstation (Qiagen) in conjunction with RNeasy 96 QIAcube HT kits (Qiagen) according to the manufacturer's recommended protocol. SMART-Seq DE3 libraries were prepared from total RNA according to the manufacturer's protocol. Briefly, for each row of the plate, polyadenylated mRNA was selected using uniquely barcoded oligo(dT) primers. First strand cDNA was generated via reverse transcription; double-stranded cDNA was created via template switching with limited cycles of PCR amplification. Each row of samples was then pooled and subjected to transposon-based fragmentation using the Nextera XT DNA Library Preparation Kit (Illumina). Libraries were then PCR amplified using unique combinations of Illumina P5 and P7 barcodes and mixed in equimolar pools prior to sequencing. Sequencing was performed on an Illumina NovaSeq6000 using custom read lengths of 89 bp (Read1) and 26 bp (Read2).

Computational workflow

Cmax annotations. Cmax values were compiled from multiple resources, and the median of these values was used to derive a consensus total Cmax. The following resources contributed to computing the consensus Cmax: Drug information from the National Center for Advancing Translational Sciences (NCATS), Porceddu et al. (2012), Khetani et al. (2013), Persson et al. (2013), Aleo et al. (2014), Garside et al. (2014), Gustafsson et al. (2014), Chen et al. (2014), Shah et al. (2015), Camenisch et al. (2019), Dixit et al. (2019), Aleo et al. (2020), Williams et al. (2020), and Smit et al. (2020). For compounds where no studies provided reliable Cmax estimates or where estimates varied significantly across studies, additional manual annotations were performed by searching PubChem and relevant clinical studies.

DILI categorization. DILIrank and LiverTox serve as key resources for categorizing drugs as DILI positive or negative. DILIrank, developed by the FDA, classifies over 1000 drugs into four levels of DILI concern – most, less, no concern, and undetermined – based on historical liver injury data. LiverTox, created by the NIDDK, is an exhaustive online database with detailed information on these drugs, including their clinical manifestations, mechanisms of action and likelihood of causing DILI. It also offers a system to assess the likelihood of DILI, from well-known cases to idiosyncratic drugs without a characteristic signature to drugs considered safe. These categorizations serve as DILI endpoints for the development and refinement of our model. Compounds were systematically categorized based on DILIrank and LiverTox into the following categories:

Withdrawn DILI: Market withdrawals and clinical failures due to DILI (Most-DILI-Concern; e.g., Troglitazone).

- *Known DILI*: Compounds with well-established DILI risk (Most-DILI-Concern in DILIrank or LiverTox score A; e.g., Isoniazid at therapeutic doses, Acetaminophen at overdose levels).
- *Likely DILI:* Compounds with documented cases of DILI in specific contexts (Most-DILI-Concern or LiverTox score A/B; e.g., Progesterone).
- *Idiosyncratic DILI:* Rare, unpredictable DILI without clear dose-response (LiverTox score C/D; <12 case reports).
- *Unlikely DILI:* Discordantly labeled safe in LiverTox (score E) but Less-DILI-concern in DILIrank.
- *No DILI:* Compounds with no clinical or preclinical documented evidence of DILI (No-DILI-Concern in DILIrank; LiverTox score E).

The categories *Withdrawn DILI*, *Known DILI*, and *Likely DILI* serve as positive controls and *No DILI* as negative control. Our models were trained to differentiate between positive and negative control compounds. The categories *Unlikely DILI* and *Idiosyncratic DILI*, due to their label ambiguity, were excluded from model training and reserved for downstream testing.

In the training data, DILI positives include the highest concentrations and those above 20x Cmax, while DILI negatives include the lowest concentrations and those below 80x Cmax. This approach ensures unambiguous labeling while minimizing bias from overdose signatures in negatives and inactive signatures in positives. Using these categorizations, our model was trained to distinguish between 177 positive and 93 negative control data points. An additional set of 70 compounds known for their idiosyncratic effects, each linked to fewer than 12 case reports, were exclusively used for testing. Idiosyncratic hepatotoxicity, characterized by its unpredictability and lack of dose-response or temporal patterns, often leads to drug withdrawals despite thorough clinical testing. An independent in-house experiment was conducted for blind validation, comprising 46 compounds: 32 DILI positives and 14 negatives. This included well-established DILI positives/negatives and a real-world set with 4 recent clinical failures.

Toxicogenomics resources and initial efforts. Human toxicogenomics benefits from resources such as TG-GATES, CMap and L1000. CMap and L1000 catalogue genomic responses in in vitro cell lines, primarily cancer lines, and can be used to study cytotoxicity, however, their utility for DILI is limited as our initial efforts showed poor predictive performance likely due to the lack of physiologically relevant cell types and relevant range of concentrations. In contrast, TG-GATES provides a rich microarray database tailored for DILI

研究，包括来自暴露于158种化合物的人类原代肝细胞（PHH）的数据。我们最初的概念验证，利用TG-GATES数据，得出了有意义的预测。在一项严格设计的内部试点实验中，涉及46种化合物进行盲法验证，我们的模型在准确识别DILI阳性方面显示了62%的敏感性，并且对阴性对照化合物的特异性为92%，显示了毒理基因组学在药物开发过程中识别DILI风险的潜力。

质量控制。我们创建了DILImap，这是我们的RNA-seq文库，涵盖了300种化合物，以提高我们机器学习模型的预测能力和机制覆盖范围。我们应用了严格的质量控制指标和筛选标准，仅保留高质量的RNA-seq样本：

1. 总RNA计数 > 平均值 −2.5 标准差（-700,000 计数）。2. 线粒体RNA比例 <9%。3. 重复样本之间的相关性 >0.99。

除了这些技术过滤器之外，我们还进行了肝细胞保真度检查：对每个样本评估肝脏特异性标记基因的表达以及肝细胞样转录特征的保留情况。

差异特征。通过所有质控（QC）过滤的样本被用于计算特定化合物的差异表达特征。我们使用 DESeq2，这是一个标准的 RNA-seq 分析工具，它采用负二项分布对读取计数进行建模。DESeq2 调整了文库大小的差异并估计了基因特异的离散度，从而能够准确检测差异表达基因（DEGs）。

对于每种化合物剂量组合，我们将处理过的样品与同一板上的匹配DMSO对照进行了比较。这种板特异性的归一化控制了潜在的批次效应，并确保所得特征反映了特定处理的转录反应。

通路特征。通路富集分析是一种统计方法，用于识别在RNA测序数据中，特定生物通路是否显著富集差异表达基因（DEGs）。该富集的显著性通过p值评估，p值表示观察到的富集发生偶然性的可能性。在我们的分析中，我们使用了广泛采用的超几何检验来计算p值。该检验计算在给定通路中观察到DEGs数量的概率，同时考虑通路中的基因总数以及数据集中DEGs的总数。为了从p值中得出通路特征，我们对调整后的p值或错误发现率（FDR）进行了-log10转换。FDR考虑了多重检验，控制分析中的假阳性比例。生成的-log10(FDR)分数作为我们模型的输入，为通路富集提供了稳健的表示。

ToxPredictor 分类器用于交叉验证。为了开发 ToxPredictor，我们评估了一系列机器学习模型，包括逻辑回归、支持向量分类器（SVC）、随机森林、梯度提升分类器、Hist 梯度提升分类器、XGBoost 分类器、LGBM 分类器和多层感知器（MLP）。每个模型都使用 5 折分层化化合物级交叉验证策略进行评估，确保来自同一化合物的所有样本在每一折中都被一起保留，以防止数据泄漏。这种化合物级的划分反映了对新化合物预测的真实泛化场景。

我们使用网格搜索优化了超参数，并评估了以下指标的性能：

• ROC曲线下面积(AUC)：衡量区分DILI和非DILI化合物的总体能力。

• 精确度：预测为 DILI 化合物中真正为 DILI 阳性的比例，与减少假阳性相关。
• 召回率：正确识别的真实 DILI 化合物的比例，反映了模型避免假阴性的能力。
• 折叠间相关性：量化在不同拆分上训练的模型之间预测概率的一致性。

• 单调剂量反应：评估化合物剂量增加是否对应预测的药物性肝损伤（DILI）风险增加。

模型选择策略

•泛化能力：优先考虑高验证AUC且训练–验证差距小，以避免过拟合 → 最佳模型的AUC为≥0.74（RF, HistGB, LGBM, XGB）；RF显示出最低的差距（-0.09 对比 GBMs 的 0.16–0.20）。

•稳定性：要求折间高度相关，以确保跨折预测的一致性 → RF 达到 0.98–1.0，高于 GBM （0.8–0.9）。

• 临床实用性：在精确率（减少假阳性）和召回率（减少假阴性）之间取得平衡 → RF的性能可与其他顶级模型相媲美。

•可解释性：在性能相近的情况下，优先选择更简单、更容易解释的模型 → 随机森林比提升方法更易解释。

最终选择：随机森林被选为最平衡且最稳健的模型，结合了强大的验证AUC、最小的过拟合、高稳定性和可解释性。尽管XGBoost和LightGBM达到了相似的AUC，但它们表现出更大的过拟合（训练-验证AUC差距）和较低的稳定性（折间相关性）。这些特性，加上其处理非线性关系的能力、对噪声的稳健性以及适用于中等规模数据集的特点，使随机森林成为对未知化合物泛化的最稳健选择。

最佳超参数为：
n_估算器 = 100, 最大_深度 = 2, 最小_样本_分割 =2, 最小_样本_叶 = 1
这种性能与稳健性的结合确立了随机森林作为ToxPredictor的基础模型。

ToxPredictor 盲验证集成建模。为了确保预测的稳健性，我们采用了由 30 个模型组成的随机森林集成，并使用自助聚合（bagging）方法。每个模型都在训练数据的独特自助样本上训练，从而保证训练实例的多样性并减轻过拟合。我们选择 30 个折叠以在准确性和集成稳定性之间取得平衡。在我们的基准测试中，由 30 个独立学习器组成的集成模型始终实现平滑、可重复的预测和稳定的剂量反应。所有折叠模型均使用交叉验证过程中优化的超参数。在对未见化合物进行预测时，模型输出采用等权重平均，生成一致性决策，从而减轻单个模型的方差，通过结合每个基础学习器的优势确保稳健表现，实现 DILI 与非 DILI 的稳定可靠分类。在盲验证中，当对未见化合物进行预测时，

DILI 风险概率。ToxPredictor 输出预测的 DILI 概率，定义为在集成的 30 个随机森林模型中的平均预测值。该概率反映了模型根据化合物的通路扰动特征判断其是否会引起肝损伤的信心。为实现下游的二元分类

research, including data from primary human hepatocytes (PHH) exposed to 158 compounds. Our initial proof-of-concept, utilizing TG-GATES data, yielded meaningful predictions. In a rigorously designed in-house pilot experiment involving 46 compounds for blind validation, our model demonstrated a 62% sensitivity in accurately identifying DILI positives and an 92% specificity for negative control compounds, demonstrating the potential of toxicogenomics in identifying DILI risks during drug development.

Quality control. We have created DILImap, our RNA-seq library encompassing 300 compounds, to improve the predictive power and mechanistic coverage of our ML model. We applied stringent quality control metrics and filtering criteria to retain only high-quality RNA-seq samples:

1. Total RNA counts > mean −2.5 standard deviations (-700,000 counts).
2. Mitochondrial RNA fraction <9%.
3. Correlation between replicates >0.99.

In addition to these technical filters, we performed a hepatocyte fidelity check: each sample was assessed for expression of liver-specific marker genes and the preservation of a hepatocyte-like transcriptional profile.

Differential signatures. Samples passing all QC filters were used to compute compound-specific differential expression signatures. We used *DESeq2*, a standard RNA-seq analysis tool, which models read counts using a negative binomial distribution. DESeq2 adjusts for library size differences and estimates gene-specific dispersion, enabling accurate detection of differentially expressed genes (DEGs).

For each compound-dose combination, we compared treated samples against matched DMSO controls from the same plate. This plate-specific normalization controls for potential batch effects and ensures that the derived signatures reflect treatment-specific transcriptional responses.

Pathway signatures. Pathway enrichment analysis is a statistical approach used to identify whether specific biological pathways are significantly enriched with differentially expressed genes (DEGs) in RNA-seq data. The significance of this enrichment is assessed using p-values, which indicate the likelihood that the observed enrichment occurred by chance. In our analysis, we utilized the widely adopted hypergeometric test to compute p-values. This test calculates the probability of observing the number of DEGs in a given pathway, considering the total number of genes in the pathway and the overall number of DEGs in the dataset. To derive pathway signatures from *p*-values, we applied a -log10 transformation of the adjusted p-value, or False Discovery Rate (FDR). The FDR accounts for multiple testing, controlling the proportion of false positives in the analysis. The resulting -log10(FDR) scores serve as the input to our model, providing a robust representation of pathway enrichment.

ToxPredictor classifier for cross-validation. To develop ToxPredictor, we evaluated a range of machine learning models, including Logistic Regression, Support Vector Classifier (SVC), Random Forest, Gradient Boosting Classifier, Hist Gradient Boosting Classifier, XGBoost Classifier, LGBM Classifier and Multi-Layer Perceptron (MLP). Each model was evaluated using a 5-fold stratified compound-level cross-validation strategy, ensuring that all samples from a single compound were held out together in each fold to prevent data leakage. This compound-level splitting reflects a realistic generalization scenario for novel compound prediction.

We optimized hyperparameters using grid search, assessing performance across the following metrics:

- **Area Under the ROC Curve (AUC):** Measures the overall ability to distinguish DILI from No-DILI compounds.
- **Precision:** Fraction of predicted DILI compounds that are truly DILI-positive, relevant for minimizing false positives.
- **Recall:** Fraction of true DILI compounds that are correctly identified, reflecting the model's ability to avoid false negatives.
- **Inter-fold correlation:** Quantifies agreement in predicted probabilities between models trained on different splits.
- **Monotonic dose response:** Assesses whether increasing doses of a compound correspond to increasing predicted DILI risk.

Model selection strategy

- **Generalizability**: prioritize high validation AUC with low train–validation gap to avoid overfitting → Best models had AUC ≥ 0.74 (RF, HistGB, LGBM, XGB); RF showed the lowest gap (-0.09 vs. 0.16–0.20 for GBMs).
- **Stability**: require high inter-fold correlation for consistent predictions across folds → RF achieved 0.98–1.0, higher than GBMs (0.8–0.9).
- **Clinical utility**: balance precision (reduce false positives) and recall (reduce false negatives) → RF performance was comparable to other top models.
- **Interpretability**: prefer simpler, easier-to-interpret models at similar performance → RF is more interpretable than boosting methods.

Final choice: *Random Forest* was selected as the most balanced and robust model, combining strong validation AUC, minimal overfitting, high stability, and interpretability. Although *XGBoost* and *LightGBM* reached similar AUCs, they showed greater overfitting (train-validation AUC gap) and lower stability (inter-fold correlation). These characteristics along with its ability to handle non-linear relationships, its robustness to noise, and its suitability for datasets of moderate size, make Random Forest the most robust choice for generalization to unseen compounds.

The optimal hyperparameters were:
`n_estimators = 100, max_depth = 2, min_samples_split = 2, min_samples_leaf = 1`

This combination of performance and robustness established Random Forest as the **base model** for ToxPredictor.

ToxPredictor ensemble modeling for blind validation. To ensure predictive robustness, we employed a 30-model Random Forest ensemble using bootstrap aggregation (bagging). Each model was trained on a unique bootstrap sample from the training data, ensuring diversity in training instances and mitigating overfitting. We selected 30 folds to strike a balance between accuracy and ensemble stability. In our benchmarks, ensembles with 30 independent learners consistently achieved smooth, reproducible predictions and stable dose-responses. All fold models used the hyperparameters optimized during cross-validation. At prediction time on unseen compounds, model outputs are averaged with equal weighting, producing a consensus decision that mitigated individual model variance, ensuring robust performance by combining the strengths of each base learner, leading to stable and reliable classification of DILI vs. No-DILI. In blind validation, where predictions were made on compounds not seen during training or model selection, the ensemble demonstrated high AUC, dose consistency, and biological plausibility, confirming its utility for real-world DILI risk assessment.

DILI risk probabilities. ToxPredictor outputs a predicted DILI probability, defined as the average prediction across the 30 Random Forest models in the ensemble. This probability reflects the model's confidence that a compound induces liver injury, based on its pathway perturbation profile. To enable binary classification for downstream

在评估和决策过程中，我们定义了0.7的DILI风险阈值。预测概率为 ≥0 ≥0.7 的化合物-剂量对被认为是DILI阳性，而预测概率为 <0<0.7 的被认为是DILI阴性。该阈值是基于经验选择的，旨在最大化我们基准数据集中敏感性和特异性之间的权衡，确保能够稳健地识别真正的DILI化合物，同时将假阳性最小化。

DILI 安全边际。ToxPredictor 会为每个化合物计算安全边际（MOS），定义为首次 DILI 剂量与治疗水平下最大血浆浓度 (Cmax) 之间的比值。首次 DILI 剂量是化合物超过 0.7 DILI 概率阈值的最低剂量，表明存在肝毒性转录证据。我们使用总 Cmax 值而非游离（未结合）Cmax，因为两种方法的预测性能相当，但总 Cmax 在临床数据来源中可获得性更广，并且与文献中常见的报告一致。

为了验证和基准测试，将一个化合物分类为DILI，我们选择了80的安全边际（MOS）阈值，该值是从训练集中经验性确定的。这个数值反映了灵敏度和特异性之间的优化权衡——在最小化假阳性的同时达到捕捉大多数真实DILI风险的性能平台。重要的是，这一阈值与文献中常用的数值一致，文献中MOS截断值通常根据具体情况在10到100之间，因此我们的选择符合公认的毒理学标准。

预测特征/途径的优先排序。为了识别和优先考虑与DILI风险预测相关的特征和途径，通过统计显著性、杂质减少、置换重要性以及直接判别力评估随机森林集成中的特征重要性，从而确保对DILI风险关键预测因子的全面且具有生物学意义的选择：

- 统计显著性：如果特征在化合物之间至少显示出一次显著的差异出现，并且p值阈值为0.001，则这些特征会被优先考虑。
- 平均不纯度减少 (MDI)：一种用于基于树的模型的标准度量，用于量化通过特定特征的分裂所实现的不纯度（例如，Gini 不纯度）减少。MDI 值在集成中的所有树上取平均，提供对特征整体预测贡献的稳健度量。具有较高 MDI 分数的特征被认为对模型的决策更为重要。
- 排列重要性：每个特征的数值被单独打乱，然后测量模型性能的变化，特别是曲线下面积（AUC）。AUC 的减小越大表示该特征的重要性越高，从而提供关于哪些特征最强烈影响模型预测的额外见解。
- 判别能力 (AUC)：通过计算每个特征的单独AUC得分，直接评估其区分DILI和非DILI类别的能力，确保具有强判别能力的特征被优先考虑。
- 斯皮尔曼相关：通路特征与预测DILI风险的相关性。

用于剂量建议的经验性 DILI 可能性。经验性累积 DILI 可能性是为了提供剂量建议，以最小化 DILI 风险而计算的。对于每种化合物，在不同的假设 Cmax 值下计算安全边际，作为模型预测的首次 DILI 剂量与相应 Cmax 的比值。对于每个安全边际，得出两个累积百分比：（1）安全边际高于给定值的 DILI 化合物的百分比，随着 Cmax 的增加从 0 单调增加到 1；（2）非 DILI 化合物的安全边际百分比低于相同的数值，其单调递减从1降至0。每个安全边际的经验累计DILI可能性通过这两个百分比的差值计算，有效地表示在给定边际处或以上，DILI化合物相对于非DILI化合物的相对富集程度。这种方法捕捉了整个数据集中安全边际与DILI可能性之间的整体关系，而不是关注孤立的点，从而能够稳健地区分高风险和低风险化合物。由此产生的累计风险曲线提供了一个定量框架，用以指导剂量选择。

基于结构的计算方法的基准测试。我们将ToxPredictor与三种最先进的基于结构的模型进行了基准测试：DILIGeNN，一种在分子图上训练的图神经网络；DILIPredictor，一种集成化学结构、物理化学性质、药代动力学参数和预测的代理DILI数据的随机森林集成模型；以及TxGemma，一种在TherapeuticData Commons (TDC)的生物医学任务上微调的通用大型语言模型(LLM)。评估旨在使用平衡准确率、敏感性和特异性来评估每种方法在重叠和未见化合物上的预测性能。

DILIPredictor：整合代理DILI标签和化学结构的随机森林集成方法。一种随机森林模型，使用九个代理DILI标签（例如线粒体毒性、BSEP抑制）结合从1111种DILI化合物的SMILES字符串中提取的化学结构特征进行训练。我们从其公共仓库复现了该模型，将Poetry环境转换为Conda（通过poetry2conda）并使用scikit-learn v1.2.0以匹配预训练模型版本。由于未提供化合物名称，我们对SMILES进行了标准化，并生成了InChIKeys（非同位素层、非立体化学层）以识别483个未见过的化合物；其中6个在推理时失败，另外6个被去重，最终得到471个（98个DILI+，373个DILI−）。如果SMILES与训练时使用的不同，这种方法仍可能引入有限的数据泄露。为了与其他计算化学模型进行基准比较，我们将重点放在314个被DILIGeNN和DILIPredictor都识别为未见过的化合物上，以提供一个逐步的

DILIGeNN：分子图上的图神经网络。一个图神经网络（GNN）在由1167个DILI化合物的SMILES字符串生成的分子图上进行训练。我们对经过连续热启动获得的性能最佳的GraphSAGE模型进行了基准测试，并使用为每个化合物推荐的自定义SMILES分子图表示。为了识别未见过的化合物，我们交叉参考了化合物名称、SMILES和InChIKey（非同位素、非立体化学层），并识别出349个未见过的化合物；其中18个推理失败，最终评估集为331个化合物（47个DILI +，284个DILI-）。其中，8个（5个DILI +，3个DILI −）与DILImap重叠，并用于基准测试。最终预测基于四个GraphSAGE模型的平均概率和多数投票。此外，我们在314个共享的未见化合物（45个DILI +，269个DILI-）上将DILI-GeNN与DILIPredictor进行了基准测试，以评估在更广泛的评价中计算机模拟模型的实际性能。

TxGemma：用于生物医学任务的通用语言模型。一套通用大型语言模型（LLMs），基于Gemma-2（2B、9B、27B参数）在治疗数据公地（Therapeutic Data Commons, TDC）的生物医学任务上进行微调。用于DILI分类，模型

evaluation and decision-making, we defined a DILI risk threshold of 0.7. Compound-dose pairs with predicted probabilities ≥0.7 are considered DILI-positive, while those <0.7 are considered DILI-negative. This threshold was empirically selected to maximize the trade-off between sensitivity and specificity in our benchmark datasets, ensuring robust identification of true DILI compounds while minimizing false positives.

DILI safety margins. ToxPredictor computes a Margin of Safety (MOS) for each compound, defined as the ratio between the first DILI dose and its maximum plasma concentration (Cmax) at therapeutic levels. The first DILI dose is the lowest dose at which a compound crosses the 0.7 DILI probability threshold, indicating transcriptional evidence of hepatotoxicity. We use *total Cmax* values rather than *free (unbound) Cmax*, as both approaches yield comparable predictive performance, but total Cmax offers broader availability across clinical data sources and aligns with common reporting in literature.

For validation and benchmarking, to classify a compound as DILI, we selected a Margin of Safety (MOS) threshold of 80, determined empirically from the training set. This value reflects an optimized trade-off between sensitivity and specificity—minimizing false positives while reaching a performance plateau that captures the majority of true DILI liabilities. Importantly, this threshold is consistent with values commonly used in the literature, where MOS cutoffs typically range from 10 to 100 depending on context, placing our choice within accepted toxicological standards.

Prioritization of predictive features/pathways. To identify and prioritize features and pathways relevant to DILI risk prediction, feature importance within the Random Forest ensemble was assessed using statistical significance, impurity reduction, permutation importance, and direct discriminative power, ensuring a comprehensive and biologically meaningful selection of key predictors of DILI risk:

- *Statistical Significance:* Features were prioritized if they demonstrated at least one significant differential occurrence across compounds, with a *p*-value threshold of 0.001.
- *Mean Decrease in Impurity (MDI):* A standard metric for tree-based models that quantifies the reduction in impurity (e.g., Gini impurity) achieved by splits on specific features. The MDI values were averaged across all trees in the ensemble, providing a robust measure of a feature's overall predictive contribution. Features with higher MDI scores were deemed more important for the model's decision-making.
- *Permutation Importance:* The values of each feature were shuffled individually, and the resulting change in model performance, specifically the Area Under the Curve (AUC), was measured. A greater reduction in AUC indicated a higher importance of the feature, offering additional insight into the features that most strongly influence the model's predictions.
- *Discriminative Power (AUC)*: Each feature's ability to distinguish between DILI and no-DILI categories was evaluated directly by calculating its individual AUC score, ensuring that features with strong discriminative capabilities were prioritized.
- *Spearman correlation:* Correlation of pathway signatures with predicted DILI risk.

Empirical DILI likelihoods for dose recommendations. Empirical cumulative DILI likelihoods are calculated to inform dose recommendations that minimize the risk of DILI. For each compound, safety margins are computed at varying hypothetical Cmax values as the ratio of the model-predicted first DILI dose to the corresponding Cmax. For each safety margin, two cumulative percentages are derived: (1) the percentage of DILI compounds with safety margins above the given value, which increases monotonically from 0 to 1 as Cmax increases, and (2) the percentage of non-DILI compounds with safety margins below the same value, which decreases monotonically from 1 to 0. The empirical cumulative DILI likelihood at each safety margin is calculated as the difference between these two percentages, effectively representing the relative enrichment of DILI compounds compared to non-DILI compounds at or above a given margin. This approach captures the overall relationship between safety margins and DILI likelihood across the dataset, rather than focusing on isolated points, thereby enabling robust differentiation between high-risk and low-risk compounds. The resulting cumulative risk profiles provide a quantitative framework to guide dose selection.

Benchmarking against structure-based in-silico methods. We benchmarked *ToxPredictor* against three state-of-the-art structure-based models: *DILIGeNN*, a graph neural network trained on molecular graphs; *DILIPredictor*, a random forest ensemble model that integrates chemical structure, physicochemical properties, pharmacokinetic parameters and predicted proxy-DILI data; and *TxGemma*, a generalist large language model (LLM) fine-tuned on biomedical tasks from the Therapeutic Data Commons (TDC). The evaluation aimed to assess the predictive performance of each approach on overlapping and unseen compounds using balanced accuracy, sensitivity, and specificity.

DILIPredictor: Random Forest ensemble integrating proxy-DILI labels and chemical structure. A Random Forest model trained on nine proxy-DILI labels (e.g., mitochondrial toxicity, BSEP inhibition) in conjunction with chemical structural features derived from SMILES strings of 1111 DILI compounds. We reproduced the model from its public repository, converting the Poetry environment to Conda (via poetry2conda) and using scikit-learn v1.2.0 to match the pretrained model version. As compound names were not provided, we standardized SMILES and generated InChIKeys (non-isotopic, non-stereochemical layer) to identify 483 unseen compounds; 6 failed at inference, 6 more were deduplicated, resulting in 471 (98 DILI + , 373 DILI −). This approach may still introduce limited data leakage if SMILES differ from those used during training. For benchmarking against other in-silico chemistry models, we focused on 314 compounds identified as unseen by both DILIGeNN and DILIPredictor, to provide a more conservative evaluation set that helps reduce the risk of data leakage; particularly important for DILIPredictor, which lacked compound identifiers and used a final model re-trained on the test set. For benchmarking against ToxPredicor, we took the overlap of the 471 compounds with DILImap yielding 30 compounds (23 DILI + , 7 DILI −).

DILIGeNN: Graph Neural Network on Molecular Graphs. A graph neural network (GNN) trained on molecular graphs derived from SMILES strings of 1167 DILI compounds. We benchmarked the best-performing GraphSAGE models obtained after sequential warm starts, and used the recommended custom molecular graph representations of SMILES for each compound. To identify unseen compounds, we cross-referenced compound names, SMILES and InChIKeys (non-isotopic, non-stereochemical layer), and identified 349 unseen compounds; 18 failed inference, yielding a final evaluation set of 331 compounds (47 DILI + , 284 DILI-). Of these, 8 (5 DILI + , 3 DILI −) overlapped with DILImap and were used for benchmarking. Final predictions were based on the mean probability and majority vote across four GraphSAGE models. In addition, we benchmarked DILI-GeNN against DILIPredictor on 314 shared unseen compounds (45 DILI + , 269 DILI-) to evaluate real-world performance of in silico models across a broader evaluation set.

TxGemma: Generalist Language Models for Biomedical Tasks. A suite of generalist large language models (LLMs), fine-tuned from Gemma-2 (2B, 9B, 27B parameters) on biomedical tasks from the Therapeutic Data Commons (TDC). For DILI classification, models

在475种化合物上进行了训练（325用于训练，54用于验证，96用于测试）。为了避免数据泄漏，我们使用SMILES和InChIKeys选择了715种未见过的化合物（178 DILI＋，537 DILI-）；然而，需要注意的是数据泄漏无法完全排除。其中，97种化合物（69 DILI＋，35 DILI-）与DILImap有重叠，并用于基准测试。TxGemma模型部署在Google Cloud Vertex AI上，使用官方手册（
h
ttps://github.com/google-gemini/gemma-cookbook/tree/main/TxGemma）和推荐的提示。27B模型表现最佳，并用于与ToxPredictor进行比较；由于模型仅提供DILI标签输出而非预测概率输出，因此未计算AUROC。通过额外的微调和提示工程，性能可能会有所提升。

本研究使用的数据分析软件。本研究使用 NumPy、pandas 和 SciPy 进行计算，使用 AnnData 处理单细胞数据，使用 s cikit-learn 1.4.0 进行机器学习，使用 matplotlib/ seaborn 进行可视化。Dask 支持可扩展计算，quilt3 管理数据集访问。DESeq2 支持差异表达分析，gseapy 支持通路富集分析。XGBoost 和 LightGB M 用于模型比较，以选择 ToxPredictor 的最佳基础模型。与计算机模拟方法的基准对比使用 PyTorch Geometric 进行图建模，Captum 用于可解释性分析，RDKit 用于化学信息学。

报告摘要
有关研究设计的更多信息，请参阅本文章所链接的《自然系列报道摘要》。

数据可用性

所有数据集和训练模型均可通过 S3 桶访问，以实现与 Jupyter 笔记本的无缝集成，并在 https://www.dilimap.org 提供详细描述和访问说明。此外，这些数据已根据知识共享许可存储在 GEO（GSE308567）中。提供了处理过的训练数据（作为模型输入使用的通路级特征）以及原始和处理过的验证数据，以实现所有模型训练和验证步骤的完全可重复性。原始训练数据（基因表达计数矩阵）包含专有信息，无法公开获取。学术研究人员可通过 DILImap@cellarity.com 申请内部非商业用途访问，请求将在 4–8 周内审核。批准的数据可使用 2 周，且必须在 6 个月内删除。商业访问需要数据共享协议。来自 Open TG-GATEs 数据库（
h
t
t
p://
dbarchive.biosciencedbc.jp/en/open-tggates/download.html）的数据。

代码可用性

所有代码、可重复性笔记本和结果可在 https://www.dilimap.org 获取，该网站作为本工作的集中访问点。完整的 Python 实现托管在 https://www.github.com/Cellarity/DILImap，带有可重复性笔记本和结果的版本在 https://www.github.com/Cellarity/DILImap_reproducibility。两个代码库也归档在 https://zenodo.org/records/17290520[61]。

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were trained on 475 compounds (325 train, 54 validation, 96 test). To avoid data leakage, we used SMILES and InChIKeys to select 715 unseen compounds (178 DILI+, 537 DILI-); however, note that data leakage could not be fully ruled out. Of these, 97 compounds (69 DILI+, 35 DILI-) overlapped with DILImap and were used for benchmarking. TxGemma models were deployed on Google Cloud Vertex AI using the official cookbook (https://github.com/google-gemini/gemma-cookbook/tree/main/TxGemma) with recommended prompts. The 27B achieved the best performance and was used for comparison with ToxPredictor; AUROC was not computed due to the model only providing DILI label outputs rather than prediction probability outputs. Performance improvements may be possible with additional fine-tuning and prompt engineering.

Data analysis software used in this study. The study used NumPy, pandas, and SciPy for computation, AnnData for single-cell data handling, scikit-learn = 1.4.0 for machine learning, and matplotlib/seaborn for visualization. Dask enabled scalable computing and quilt3 managed dataset access. DESeq2 supported differential expression and gseapy pathway enrichment. XGBoost and LightGBM were used for model comparison to select the optimal base model for *ToxPredictor*. Benchmarking against in-silico methods employed PyTorch Geometric for graph modeling, Captum for interpretability, and RDKit for cheminformatics.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets and trained models are accessible via an S3 bucket for seamless integration with Jupyter notebooks, with detailed descriptions and access instructions provided at https://www.dilimap.org. In addition, the data have been deposited in GEO under a Creative Commons license (GSE308567). Processed training data (pathway-level signatures used as model input) and both raw and processed validation data are provided to enable full reproducibility of all model training and validation steps. The raw training data (gene expression count matrices) contain proprietary information and are not publicly available. Academic researchers may request access for internal, non-commercial use via DILImap@cellarity.com, with requests reviewed within 4–8 weeks. Approved data are available for 2 weeks and must be deleted within 6 months. Commercial access requires a data-sharing agreement. The data from the Open TG-GATEs database (http://dbarchive.biosciencedbc.jp/en/open-tggates/download.html)[15] were used in an initial proof-of-concept to establish a toxicogenomics baseline performance, whereas all results reported in this manuscript are based on the internally generated data described above. The use of all datasets in this study complies with the terms and conditions of their respective repositories and data providers.

Code availability

All code, reproducibility notebooks, and results are available at https://www.dilimap.org, which serves as the central access point for this work. The full Python implementation is hosted at https://www.github.com/Cellarity/DILImap, with reproducibility notebooks and results at https://www.github.com/Cellarity/DILImap_reproducibility. Both repositories are also archived at https://zenodo.org/records/17290520[61].

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致谢

我们衷心感谢比尔·佩尼（Bill Pennie）和阿特利·索拉伦森（Atli Thorarensen）提出了塑造本工作的关键想法。我们感谢戈温达·比塞蒂（Govinda Bhisetti）对手稿进行的全面审阅。我们的感谢还送给罗布·尼斯旺格（Robb Nicewonger）在化学操作规程方面的工作，卡梅隆·赖利（Cameron Reilly）在RNA提取和自动化方面的支持，劳拉·伊萨科（Laura Isacco）在协调数据生成时间表方面的工作，邵·陈（Thao Tran）和温特·盖斯（Wynter Guess）在化合物筛选方面的协助，温妮·李（Winnie Lee）和布赖恩·易（Brian Yi）在RNA提取方面的工作，所有这些都为本工作的数据生成奠定了基础。最后，我们感谢审稿人提出的深思熟虑且具有建设性的建议，这些建议帮助完善和加强了本工作。

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Acknowledgements

We sincerely thank Bill Pennie and Atli Thorarensen for bringing in pivotal ideas that shaped the work. We appreciate Govinda Bhisetti for the thorough review of the manuscript. Our gratitude extends to Robb Nicewonger for his work on the chemistry SOP, Cameron Reilly for his support with RNA extraction and automation, Laura Isacco for coordinating data generation timelines, Thao Tran and Wynter Guess for their assistance with compound screening, Winnie Lee and Brian Yi for their work on RNA extraction, all of which contributed to data generation foundational to this work. Finally, we thank the reviewers for their thoughtful and constructive requests, which helped refine and strengthen the work.

作者贡献
V.B. 和 K.K. 构思并共同领导了该项目。V.B. 开发了框架并撰写了手稿。K.K. 指导了细胞培养实验的设计和执行。S.S. 实施并进行了与最先进的计算模型的基准测试。O.B. 对研究的概念框架做出了贡献。S.A. 参与了培养系统的开发并进行了活性筛查。M.R.

建立了初步实验并参与了RNA-seq方法的选择。C.F. 实现了数据生成工作流程的自动化。H.B. 负责化合物管理、筛选工作并参与了化学标准操作程序（SOP）的制定。N.R. 执行了测序用文库的构建。T.F. 参与了化合物管理和筛选工作。N.L. 建立了Dotmatics工作流程。M.C. 监督了数据生成活动。N.P. 参与了最初使用转录组学建模非靶向效应的概念设计。M.G. 提供了战略方向。M.Z. 对机器学习方面以及战略方向提供了监督。

利益冲突
所有作者在进行本研究时均为Cellarity Inc.的员工，并持有公司股权。该研究由Cellarity资助。V.B.、K.K. 和 O.B. 是专利申请 WO2025/024525 的发明人，该专利涉及DILI预测方法，归Cellarity所有。

附加信息
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同行评审信息 《自然通讯》感谢 Volker Lauschke、James Dear 以及其他匿名审稿人在本工作的同行评审中所作的贡献。同行评审文件可用。

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Author contributions
V.B. and K.K. conceived and co-led the project. V.B. developed the framework and authored the manuscript. K.K. directed the design and execution of the cell culture experiments. S.S. implemented and ran benchmarking against state-of-the-art in silico models. O.B. contributed to the study's conceptual framework. S.A. contributed to the development of the culture system and conducted viability screens. M.R. established the initial experiments and contributed to RNA-seq method selection. C.F. implemented automation for data generation workflows. H.B. led compound management, screening and contributed to the chemistry SOP. N.R. performed library construction for sequencing. T.F. contributed to compound management and screening. N.L. established the Dotmatics workflows. M.C. supervised the data generation activities. N.P. was involved in the original conception of using transcriptomics to model off-target effects. M.G. provided strategic direction. M.Z. provided oversight for the machine learning aspects and strategic direction.

Competing interests
All authors were employees of Cellarity Inc. at the time the study was performed and hold equity in the company. The research was funded by Cellarity. V.B., K.K., and O.B. are inventors on patent application WO2025/024525, related to methods for DILI prediction, assigned to Cellarity.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-025-65690-3.

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Peer review information *Nature Communications* thanks Volker Lauschke, James Dear and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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